Highlights

Auditing the Decision Value of Physics-Constrained Wind Power Forecasting

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- Systematic scoping audit of 46 physics-constrained wind power forecasting studies.
- Literature audit reveals 95.7% open-code deficit and zero documented negative trade-offs.
- Software audit of PCT identifies reverse-argument JS divergence altering gradient scales.
- Scrambled priors induce a +68.1% decision loss surge under strong constraints on Penmanshiel.
- Fixed historical priors assist limited training windows but diminish under sample abundance.
- Degenerate zero-bid probe exposes the methodological hazards of unhedged cashflow evaluation.

Auditing the Decision Value of Physics-Constrained Wind Power Forecasting

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Abstract

Reductions in point forecasting error from physical constraints do not establish that empirical gains stem from valid physical information, nor do they guarantee operational decision value. We audit the attributional foundation of physics-constrained wind power forecasting through systematic literature scoping, software code inspection, and multi-site controlled experiments across three commercial UK wind farms. Among 46 scoped studies (2023–2026), zero report operational failure regimes or trade-offs, 44 lack accessible source code, and 23 evaluate constraints by excising multi-layer network modules, confounding physical inductive bias with parametric capacity. In controlled ablations holding the forecasting backbone and post-hoc quantile calibration constant, the impact of empirical power curve priors varies across data regimes: on Penmanshiel, strong constraints reduce normalized asymmetric decision loss by 1.22%, whereas destroying the aerodynamic mapping via pseudo-scrambled controls inflates decision loss by 68.1%, demonstrating prior specificity under data scarcity. Across well-sampled sites, fixed historical priors provide marginal loss reductions under constrained training windows, an advantage that diminishes and reverses as model training sample sizes expand. Software code inspection reveals an inverted parameter argument in an official divergence loss implementation that alters gradient scales, while market settlement diagnostics demonstrate that standalone unhedged balancing cashflows decouple from forecast quality under degenerate zero-bid bidding. These findings indicate that empirical claims of physical benefit require backbone-matched negative controls, transparent disclosure of prior historical data budgets, and consistent decision-theoretic evaluation.

Keywords: Physics-Constrained Deep Learning, Wind Power Forecasting, Prediction-Decision Decoupling, Conditional Inductive Priors, Systematic Audit, Implementation Fidelity, Machine Learning Reproducibility

1. Introduction

In electric power systems and spot markets, the operational value of wind power forecasting is ultimately determined by asymmetric decision losses in day-ahead commitment and balancing dispatch [1, 2]. Under dual-settlement balancing rules, under-delivery penalties typically diverge sharply from over-delivery spill revenues. In these asymmetric operational regimes, conventional symmetric regression criteria such as Root Mean Squared Error (RMSE) or Mean Absolute Error (MAE) fail to represent downside financial and system reliability exposure [11]. To improve generalization and prevent physically anomalous outputs, a surging body of literature embeds aerodynamic limits, empirical turbine power curves, or kinematic equations directly into deep neural network architectures and loss functions [7, 6, 8, 10]. A foundational premise underpins this research paradigm: that reported reductions in forecasting error can be directly attributed to the valid domain knowledge encoded within the physical constraint.

However, prevailing reporting conventions, software availability deficits, and ablation designs obscure this attributional premise. First, across contemporary literature, published studies predominantly document positive outcomes; empirical failures, negative trade-offs, or constraint breakdowns remain virtually absent. Second, public source code availability remains exceptionally rare, leaving computational graph executions unverified [12, 13]. Third, where ablation studies are reported, they frequently excise entire multi-layer neural submodules (such as dual-channel encoders or dynamic attention blocks) rather than isolating the physical penalty parameter holding model capacity constant. Such coarse ablations inevitably confound aerodynamic inductive bias with parametric representation capacity. Consequently, the literature leaves fundamental questions unresolved:

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does a physical penalty succeed because the network internalizes genuine aerodynamic relationships, or because the penalty merely acts as a non-specific numerical regularizer? Under what training sample budgets, temporal proxy lags, and evaluation metrics do these empirical advantages persist, degrade, or invert?

To examine the attributional basis of physics-constrained gains, we conduct a controlled empirical audit across three commercial UK wind farms (Kelmarsh, Hill of Towie, and Penmanshiel). Holding the forecasting backbone (DLinear) and post-hoc asymmetric quantile calibration strictly identical, we introduce scale-matched pseudo-scrambled priors as negative controls, systematically perturb prior parameters, and vary model training horizons. The controlled experiments reveal that physical constraint advantages are bounded and contingent:

1. **Prior Specificity vs. Misspecification Fragility**: On Penmanshiel under restricted training samples (N_{train} = 1,583), a strong power curve constraint reduces normalized asymmetric decision loss by 1.22%. However, permuting the wind-speed-to-power mapping while preserving loss formulation and gradient magnitude inflates decision loss by 68.1%. This contrast confirms that while valid domain information can provide measurable inductive advantage in data-scarce settings, it introduces substantial vulnerability when the prior is misspecified.

2. **Information Budgets and Sample Regimes**: When models are evaluated under varying training sample budgets, fixed historical empirical priors provide modest benefits under limited training windows (e.g., N = 500, representing ∼ 21 days of recent operations). However, because the prior is estimated from the full historical training span (7 to 10 months), it injects significant additional information. As the model's own training window expands to N = 5,000, this prior advantage diminishes and eventually reverses, where rigid historical curves constrain the model from fitting shifting local dynamics.

3. **Implementation Divergence and Market Metric Decoupling**: At the software execution level, an inspection of open-source computational graphs identifies an inverted parameter argument in the Jensen–Shannon divergence loss of the Physics-Constrained Transformer (PCT), which induces asymmetric gradient amplification. Downstream, an analysis of market settlement demonstrates that standalone unhedged balancing cashflow is structurally decoupled from forecasting skill: a degenerate zero-bid strategy ($\hat{y} \equiv 0$) mechanically accrues substantial apparent cashflow despite generating zero predictive information.

1.1. Summary of Contributions

This audit addresses the attribution and evaluation of physics-constrained forecasting through three interconnected contributions:

- **Scoping Audit of Literature Reporting and Code Availability**: We evaluate 46 contemporary studies (2023–2026) under PRISMA-ScR guidelines, quantifying the reporting absence of operational failure regimes (0/46, 0.0%), the deficit of open-source artifacts (44/46, 95.7%), and the prevalence of structural module-level ablation confounding (23/46, 50.0%).

- **Backbone-Matched Controlled Empirical Ablation**: By pairing authentic priors with scrambled negative controls across 3–4 independent seeds per wind farm on a dedicated 10-GPU cluster, we isolate domain-specific inductive bias from numerical smoothing. We delineate the operational boundaries under which physical constraints assist, become indistinguishable from random regularization, or degrade decision performance.

- **Computational Graph and Settlement Decoupling Diagnostics**: We uncover implementation divergence in public constraint software and mathematically resolve why unhedged single-settlement imbalance cashflows fail as a surrogate for forecast quality, establishing a decision-theoretic audit framework anchored to normalized asymmetric dispatch loss.

Empirical Scope and Methodological Boundaries. This investigation defines four explicit operational boundaries: first, the scoping literature audit evaluates reporting and ablation practices across the 46 contemporary studies identified under our PRISMA-ScR protocol (38 peer-reviewed journal articles and 8 preprints, 2023–2026), characterizing reporting gaps within this scoping cohort rather than universal disciplinary properties; second, the software audit examines the computational graph of the open-source PCT implementation (commit `65941ec`); third, the empirical ablations characterize boundary behavior across three commercial UK wind farms under matched DLinear backbones with PatchTST serving as an architectural sensitivity probe; and fourth, the downstream evaluations conduct an operational decision-risk and single-settlement imbalance cashflow diagnostic under asymmetric two-settlement rules, establishing metric failure modes rather than full commercial portfolio clearing. Together, these four components establish an

evidence-grounded assessment from mathematical specification to market dispatch.

2. Related Work and Positioning

To position this scoping empirical audit within computational energy systems and empirical machine learning methodology, we examine three intersecting areas: (i) physics-constrained formulations in renewable energy forecasting, (ii) benchmarking and operational decision value in wind power prediction, and (iii) reproducibility, reporting checklists, and software auditing in empirical AI.

2.1. Three Methodological Dilemmas in Physics-Constrained Energy Forecasting

The integration of physical principles into neural estimators is widely presented as a synthesis of domain governing equations and data-driven learning [6, 7]. In computational renewable energy systems, this formulation encounters three structural dilemmas:

Dilemma 1: Kinematic Hard Clipping versus Turbulent Manifolds. Enforcing aerodynamic operational limits (cut-in, rated, and cut-out wind speeds, or the theoretical Betz envelope $P \leq \frac{16}{27} \cdot \frac{1}{2}\rho A v^3$) via hard bounds or bounded activations guarantees output admissibility [9]. However, hard clipping alters high-order gradient geometry across saturation boundaries. When atmospheric boundary-layer turbulence transitions turbine operation into non-stationary regimes, zero-gradient saturation zones stall backpropagation, truncating the neural manifold and preventing adaptation to transient aerodynamic excursions.

Dilemma 2: Spatio-Temporal Wake Graph Networks versus Low-Order Empirical Algebraic Matrices. Embedding spatial wake deficits—such as 1D kinematic Jensen or Bastankhah–Porté-Agel Gaussian formulations [8, 15]—into Graph Neural Network (GNN) adjacency matrices substitutes empirical steady-state engineering approximations for dynamic Navier–Stokes continuum physics. Imposing 1D scalar algebraic matrices onto high-capacity spatiotemporal graph representations restricts the network from resolving time-varying turbulent eddy propagation. Published evaluations frequently attribute downstream relational gains to the embedded engineering surrogate rather than testing whether the expressive capacity of the GNN backbone alone accounts for the improvement.

Dilemma 3: Distribution Divergence Regularization versus Loss Landscape Stiffness. Penalizing deviations from historical joint wind speed–power densities via statistical divergence penalties (Wasserstein distance or Jensen–Shannon divergence [10, 9]) introduces severe loss landscape stiffness [16]. Static distribution alignment projects historical average manifolds onto non-stationary meteorological time series. When runtime telemetry exhibits observation latency or atmospheric regimes shift, multi-objective penalty gradients penalize physically valid operational transients, inducing runtime proxy drift and inflating downstream dispatch imbalance costs.

2.2. Benchmarking, Robustness, and Decision Value in Wind Forecasting

Evaluation of short-term wind power forecasting has long relied on standard statistical error metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) [17, 18]. Madsen et al. [17] established standardized protocols for short-term wind power evaluation, highlighting the necessity of assessing models across distinct operational regimes, forecast horizons, and normalized capacity metrics. More recently, Optis et al. introduced the open-source *WE-Validate* framework [18], advocating for resistant and distributionally robust metrics that reflect meteorological variability and complex terrain dynamics.

Power systems engineering demonstrates that benchmark accuracy on historical test splits diverges from operational value under asymmetric grid penalties [2, 19]. Pinson et al. [20] proved that optimal electricity market bidding is governed by conditional quantiles matching asymmetric imbalance penalties, establishing the operational separation between point accuracy and financial risk. Grid operations penalize reserve shortfalls and unhedged ramp errors far more severely than symmetric variance [21, 22]. Real-world balancing mechanisms, such as the UK Elexon Balancing and Settlement Code (BSC), evaluate net physical imbalances over discrete 30-minute settlement periods, where positive and negative deviations net out algebraically before incurring cashout charges. Yet published studies routinely evaluate models on unhedged single-settlement cashflows, mistaking accidental prediction bias for forecasting skill. Evaluating neural forecasters strictly on uncalibrated point metrics or unhedged cashflows obscures their true operational reliability.

2.3. Reproducibility, Reporting Checklists, and Software Auditing in Machine Learning

Empirical machine learning has faced increasing scrutiny over data leakage, underspecification, and unverified software implementations [24, 23, 12, 11]. Kapoor and Narayanan [12] documented reproducibility failures stemming from subtle leakage vectors across seventeen scientific disciplines, showing that data-driven claims frequently collapse when subjected to rigorous independent re-examination. Gundersen and Kjensmo [23] conducted an empirical audit of 400 papers presented at AI venues (AAAI and IJCAI), finding that fewer than 20% provided accessible code or reproducible experimental protocols.

To address these vulnerabilities, leading scientific venues have introduced procedural interventions. Pineau et al. [13] developed the NeurIPS Machine Learning Reproducibility Checklist, demonstrating that formal reporting mandates correlate with improved disclosure of code, data, hyperparameter tuning, and error bounds. Beyond documentation, D'Amour et al. [24] demonstrated that models exhibiting equivalent in-distribution benchmark accuracy often diverge sharply under operational stress due to underspecified training objectives. Concurrently, Lipton and Steinhardt [11] identified pervasive weaknesses in empirical scholarship: conflating empirical correlation with theoretical causation, failing to isolate independent sources of empirical gain, and deploying mathematical formalism decoratively without operational consequences.

While general machine learning venues have increasingly adopted these auditing norms, physics-constrained energy forecasting has developed largely without standardized reporting controls. As summarized in Table 1, existing reviews focus primarily on architectural taxonomy and qualitative typologies, rarely probing constraint provenance, software parity, or the causal validity of ablation studies. This work operationalizes these empirical standards through systematic auditing across constraint lifecycle provenance, software parity, and controlled ablations.

3. Conceptual Framework: The Five-Dimensional Constraint Lifecycle (Schema v4)

In renewable energy forecasting, designating a neural network as "physics-constrained" conflates disparate mathematical, structural, and empirical operations under an umbrella descriptor [11]. To establish an operational foundation for empirical auditing, we introduce the **Constraint Lifecycle Taxonomy (Schema v4)**.

3.1. Unit of Analysis and Lifecycle Representation

In our scoping framework, the primary unit of analysis is the individual **forecasting study** ($N = 46$ unique scoping studies), characterized by its primary operational constraint mechanism. For forecasting methodologies that integrate multiple physical priors (e.g., combining an empirical power curve regularizer with a spatial wake graph architecture), each study is classified according to its dominant physical constraint governing the forecasting computational graph, while multi-component interactions and secondary constraints are preserved in the open audit ledger. Formally, we characterize the primary constraint architecture C of each audited study as a 5-tuple operating across orthogonal lifecycle stages:

$$C = \langle \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4, \mathcal{D}_5 \rangle, \tag{1}$$

where $\mathcal{D}_1$ represents parameter provenance, $\mathcal{D}_2$ denotes application stage, $\mathcal{D}_3$ specifies test-time inference telemetry requirements, $\mathcal{D}_4$ captures implementation software fidelity, and $\mathcal{D}_5$ evaluates harm reporting and ablation validity. The overarching lifecycle trajectory from physical specification to operational deployment is illustrated schematically in Figure 1.

3.2. Operational Definitions of the Five Lifecycle Dimensions

Schema v4 characterizes each audited constraint architecture through five orthogonal lifecycle dimensions:

Dimension 1: Parameter Provenance ($\mathcal{D}_1$). Parameter provenance identifies the epistemic origin and empirical calibration of the numerical values governing a constraint:

- *Intrinsic constraints*: parameters derive analytically from governing physical invariants, manufacturer nameplate limits, or classical equations without empirical fitting (e.g., the Betz limit $C_p \leq 16/27$ or kinematic wake equations).
- *Historical-static priors*: parameters or reference distributions are estimated offline from historical SCADA archives and held fixed during inference (e.g., empirical power curves fitted via kernel density estimation).
- *Learned-online constraints*: parameters adapt dynamically during test-time inference via an auxiliary neural subnetwork.

When a study omits calibration details, it is designated as *unreported*. Distinguishing provenance is critical: static empirical priors implicitly assume aerodynamic stationarity, risking silent miscalibration under turbine aging or curtailment.

Table 1: Positioning this Scoping Empirical Audit Relative to Existing Literature Reviews in Wind Power Forecasting and Scientific ML.

Survey / Meta-Review Literature Reference	Decouples Provenance ($\mathcal{D}_1$)	Audits Stage Separation ($\mathcal{D}_2$)	Checks Inference Telemetry ($\mathcal{D}_3$)	Verifies Code Parity ($\mathcal{D}_4$)	Scrutinizes Ablation ($\mathcal{D}_5$)
Yan et al. (2016) [25]	Partial	No	Partial	No	No
Maldonado-Correa et al. (2020) [26]	No	No	No	No	No
Karniadakis et al. (2021) [6]	Conceptual	Conceptual	No	No	No
Optis et al. (WE-Validate, 2022) [18]	No	No	Partial	Open Benchmark	No
Sheng et al. (2025) [8]	No	Partial	No	No	No
This Scoping Empirical Audit	**Yes (Schema v4)**	**Yes (Schema v4)**	**Yes (Schema v4)**	**Yes (Static Audit)**	**Yes (Strict Isolation)**

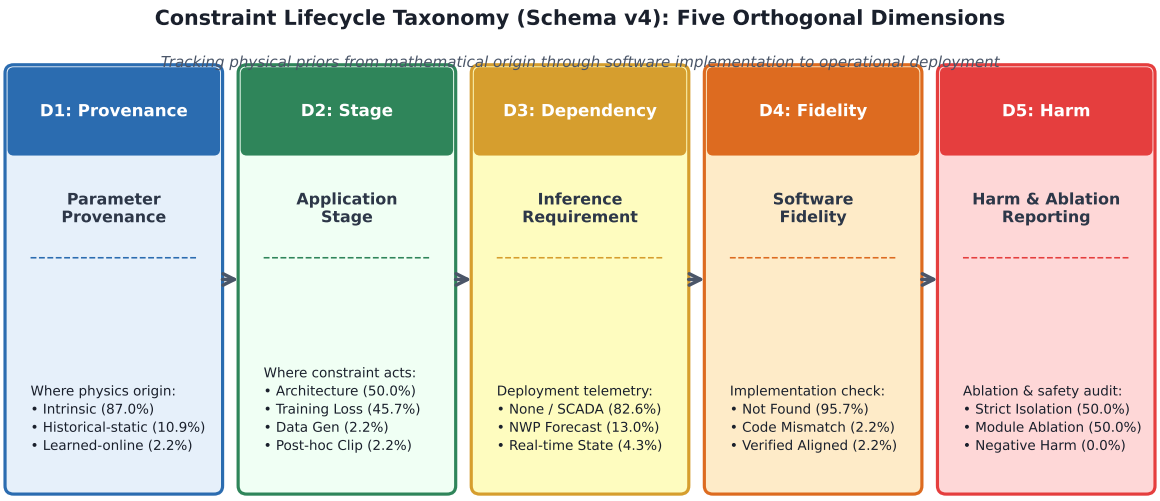


Figure 1: The Five-Dimensional Constraint Lifecycle Framework (Schema v4). Tracking domain physical constraints from their mathematical origin ($\mathcal{D}_1$) through computational graph formulation ($\mathcal{D}_2$), operational telemetry dependency ($\mathcal{D}_3$), software implementation parity ($\mathcal{D}_4$), and empirical safety/ablation reporting ($\mathcal{D}_5$).

Dimension 2: Application Stage ($\mathcal{D}_2$). This dimension identifies the computational point where domain physics enters the learning pipeline: (1) *training-loss regularization*, where physics acts as a differentiable penalty appended to the loss objective ($\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda\mathcal{L}_{\text{phys}}$); (2) *inference-architecture constraints*, which modify the computational graph to project outputs onto feasible manifolds; (3) *data-generation priors*, used upstream to synthesize or augment training sequences; and (4) *post-hoc correction filters*, applied as external mathematical projections after neural inference.

Dimension 3: Inference Telemetry Requirements ($\mathcal{D}_3$). Inference telemetry identifies external data streams required to evaluate constraints during real-time deployment: (1) *none*, relying solely on target turbine historical power series; (2) *forecast-input*, strictly mandating future meteorological inputs (e.g., forward numerical weather prediction wind vectors); and (3) *contemporaneous-state*, requiring synchronized, zero-latency measure-

ments from adjacent turbines or met masts. Classification as $\mathcal{D}_3$ = `none` indicates that the base forecasting model requires no external runtime telemetry streams (such as real-time meteorological feeds or forward NWP data). When test-time physical constraint evaluation relies on external meteorological proxies (such as querying an empirical power curve or joint distribution prior with wind speed), operators lacking forward NWP or zero-latency SCADA feeds must either omit the constraint or substitute lagged historical observations. Observation staleness is therefore an operational hazard specific to proxy-dependent constraint architectures under telemetry delays, rather than an automatic attribute of $\mathcal{D}_3$ = `none` models.

Dimension 4: Implementation Fidelity ($\mathcal{D}_4$). Implementation fidelity evaluates whether public software faithfully executes the published mathematical formulations. Codebases are classified as: (1) *verified-aligned*, where public source code matches published equations; (2) *code-mismatch*, where code exhibits an algebraic or directional deviation from the text; and (3) *not-found*, where no public repository or script accompanies the paper. Mathematical typesetting often conceals divergence in numerical execution, making independent software auditing essential.

Dimension 5: Harm Reporting and Ablation Semantics ($\mathcal{D}_5$). This dimension examines whether evaluations causally isolate physical constraints and report operational failure boundaries. Evaluation protocols are categorized into: (1) *strict-isolation*, where treatment and baseline share identical parameter capacity, differing only in the activation of the physical prior ($\lambda = 0$ vs. $\lambda > 0$); (2) *module-ablation*, where the ablation excises an entire multi-parameter neural block, confounding structural capacity with domain knowledge; (3) *both*, reporting both isolated toggles and architectural excisions; and (4) *harm-reported*, explicitly documenting operating regimes where constraints degrade performance.

3.3. Methodological Principles for Schema Disambiguation

To ensure reproducible coding across independent raters, Schema v4 enforces four disambiguation rules. Parameter provenance ($\mathcal{D}_1$) separates offline calibration from runtime telemetry ($\mathcal{D}_3$), ensuring that historical estimation methods are not conflated with active SCADA stream dependencies. Application stage ($\mathcal{D}_2$) is determined by the backpropagation path: loss regularizers update network weights through objective penalties during training, whereas inference-architecture modifications constrain the forward computational graph directly under arbitrary parameter states. For composite methodologies that incorporate secondary constraints alongside structural modifications (e.g., both graph topology and auxiliary loss penalties), the primary classification is determined by the structural constraint directly modifying the forward computational graph, with secondary mechanisms logged in the qualitative audit ledger. Causal ablation validity ($\mathcal{D}_5$) requires parameter-matched controls; excising multi-layer neural submodules evaluates overall architectural capacity rather than the specific inductive contribution of the physical prior.

4. Scoping Audit Protocol and Empirical Test Bench

We conducted an empirical scoping audit in accordance with the PRISMA Extension for Scoping Reviews (PRISMA-ScR) guidelines [27] and established scoping methodology [28]. The audit protocol was specified prior to full-text extraction to govern retrieval strings, eligibility thresholds, multi-pass screening, and inter-rater deduplication.

4.1. Information Sources and Retrieval Strategy

We retrieved candidate literature across two complementary discovery channels covering January 1, 2023, through March 1, 2026, with rolling monitoring through July 2026 for emergent preprints. First, we conducted programmatic searches via the OpenAlex API, which indexes records across IEEE Xplore, ScienceDirect, SpringerLink, Wiley, and MDPI. Using four targeted title queries—Q_1 (`"physics-constrained" AND "wind"`), Q_2 (`"physics-informed" AND "wind power"`), Q_3 (`"physics-guided" AND "wind"`), and Q_4 (`"physics-embedded" AND "wind power"`)—we identified 72 candidate records. Second, we conducted targeted searches on Google Scholar, arXiv, and preprint servers to capture early-access journal articles and working papers, yielding 17 additional records ($N = 89$ total). Candidate records were cross-checked against Scopus and IEEE Xplore to verify coverage of major power systems and AI venues. All search strings, raw records, and exclusion criteria are documented in our open supplementary ledger. Furthermore, a targeted sensitivity search expanding retrieval to title, abstract, and author keywords across Scopus and IEEE Xplore (encompassing broader descriptors such as `"knowledge-guided"`, `"hybrid"`, and `"domain-informed"`) identified 14

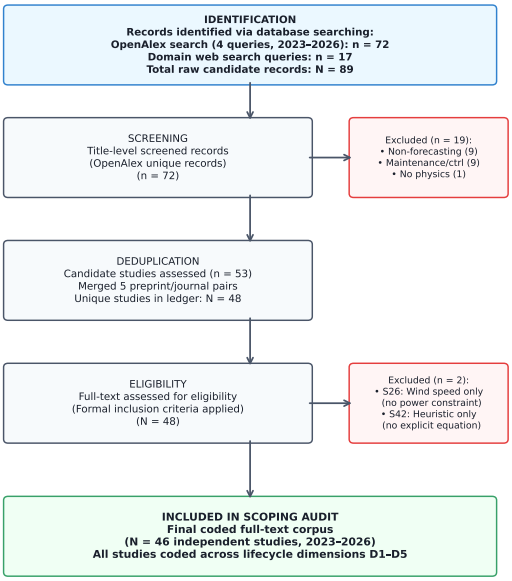


Figure 2: PRISMA-ScR Flow Diagram Illustrating the Scoping Corpus Derivation (89 → 72 → 53 → 48 → 46). Deduplication merged 5 preprint/journal pairs, while full-text eligibility screening excluded 2 studies, leaving 46 eligible full-text studies.

additional papers. Full-text review confirmed that none documented an operational failure boundary or negative trade-off, specifically verifying that the 0.0% negative-harm prevalence finding is robust to broader search scope. We note that this targeted 14-study query served strictly to stress-test negative trade-off reporting; exhaustive five-dimensional extraction ($\mathcal{D}_1$–$\mathcal{D}_5$) and inter-rater reliability coding were conducted on the core PRISMA-ScR scoping cohort of 46 studies.

4.2. Screening, Deduplication, and Eligibility Criteria

Candidate records underwent structured multi-phase screening following the PRISMA-ScR workflow (Figure 2). Cross-source deduplication first eliminated 17 duplicate records from the initial 89 items, leaving 72 unique entries for title and abstract evaluation. Screening at this stage excluded 19 records with documented rationale: nine addressed non-forecasting tasks such as turbine structural health monitoring, blade crack detection, or aeroelastic fatigue modeling; nine focused on real-time blade pitch or active power control rather than multi-step power forecasting; and one contained no physics-based modeling components. This filtering yielded 53 included candidate studies sought for full-text retrieval.

Because authors frequently release preliminary working papers on preprint servers (such as arXiv or SSRN) prior to formal journal publication, treating preprints and published articles independently would artificially inflate corpus statistics. We enforced a strict study-level deduplication rule: whenever a preprint and a peer-reviewed journal paper shared identical authorship, empirical datasets, and mathematical formulations, they were merged into a single study-level record retaining the peer-reviewed journal citation. Five paired duplicate sets were identified and consolidated (e.g., *Applied Energy* 2025 paired with SSRN 2024, and *Electric Power Systems Research* 2026 paired with SSRN 2026), reducing the candidate cohort from 53 to 48 unique study-level entries.

The remaining 48 full texts were independently evaluated against three explicit scoping eligibility criteria: (1) primary focus on short-term or ultra-short-term wind power forecasting; (2) mathematical formulation and computational execution of an explicit physical constraint, governing equation, or domain envelope; and (3) empirical validation on real-world or simulated wind power generation time series. Two studies failed to meet these criteria: study S26 was excluded because it forecasted atmospheric wind velocity exclusively without an explicit turbine power constraint, while study S42 was excluded because it implemented heuristic feature clustering without domain physical equations. This systematic filtering established our final representative scoping corpus of $N = 46$ **unique scoping studies**, comprising 38 peer-reviewed journal articles and eight representative preprints (including UniWind [29]).

4.3. Multi-Rater Reliability Benchmarking

To assess the empirical stability of the Schema v4 taxonomy before full ledger extraction, three independent raters (Coder A, Coder B, and Coder C) double-blindly coded a stratified representative subset of 20 candidate studies across all five dimensions. The pilot benchmarking achieved substantial to perfect inter-rater concordance across taxonomy facets. Study eligibility screening achieved unanimous consensus (Fleiss $\kappa = 1.000$), correctly isolating negative controls S26 and S42. Parameter provenance ($\mathcal{D}_1$) demonstrated perfect agreement ($\kappa = 1.000$ across both total and included corpora), while application stage ($\mathcal{D}_2$) yielded near-perfect agreement ($\kappa = 0.950$ overall, $\kappa = 0.938$ on included studies). Inference telemetry requirements ($\mathcal{D}_3$) achieved substantial agreement ($\kappa = 0.743$ overall, $\kappa = 0.646$ on included studies).

In contrast, implementation fidelity ($\mathcal{D}_4$) exhibited moderate concordance prior to consensus arbitration ($\kappa = 0.716$ overall in the stratified $N = 20$ pilot with negative controls, but $\kappa = 0.341$ when evaluated strictly on included studies). This reduction reflects the classical prevalence paradox described by Feinstein and Cicchetti: because 92.6% of candidate studies omitted public code, expected chance agreement P_e was high (86.1%), which depresses Fleiss' κ despite a raw observed agreement of $P_o = 90.7\%$. Evaluating static manuscripts without code repositories leaves coders to interpret ambiguous narrative descriptions regarding whether a constraint represents an active computational component or conceptual motivation. This divergence indicates that software implementation fidelity cannot be reliably inferred from published prose alone, motivating direct computational graph inspection. For ablation semantics ($\mathcal{D}_5$), raters achieved strong concordance ($\kappa = 0.851$ overall, $\kappa = 0.818$ on included studies). On harm reporting, all three raters unanimously coded zero instances of harm documentation across the 18 included studies (0/18); because marginal variance was zero ($P_e = 1.0$), Cohen's κ is mathematically indeterminate (0/0) and is formally reported as 100% observed agreement ($P_o = 1.000$). Following pilot calibration and consensus arbitration, full-text extraction across the complete 46-study corpus was conducted under Schema v4 at the study level (coding each study's primary operational constraint mechanism alongside verbatim textual citations and equation coordinates), with primary extraction executed by J.L. and cross-verification of key quantitative parameters conducted by J.D.

4.4. Software Audit Protocol (PCT & PINN-UQ)

The code availability audit followed a standardized five-step software search protocol examining full-text bodies, footnotes, data availability statements, publisher repositories, and author profiles. Across the 46-study corpus, only two publicly accessible software repositories were identified. One repository, PINN-UQ (S10, *Engineering Applications of Artificial Intelligence* 2025), was audited and verified to faithfully implement its specified algebraic power equations ($P = gT\omega$ and $P = \frac{1}{2}C_p\rho Av^3$ via `sn.Tie`); its computational graph strictly executes the published algebraic electromechanical relationships without unstated differential substitutions, earning classification as *verified-aligned*. The remaining repository, the Physics-Constrained Transformer (PCT, S02, *Scientific Reports* 2026), provided open PyTorch scripts (`train_clean.py` and `tst/loss.py`) which we audited line-by-line for parity between published loss formulations and computational graph execution, as detailed in Section 6.

4.5. Downstream Operational Dispatch and Settlement Test Bench

To ground the empirical audit in real-world power system operations across diverse meteorological and topographic terrains, we established an operational dispatch test bench across three commercial UK wind farms with 10-minute SCADA telemetry. The *Hill of Towie Wind Farm* (48.3 MW rated capacity, 21 Siemens SWT-2.3-82 turbines) features complex upland terrain and serves as an independent spatiotemporal extrapolation benchmark (7,599 test origins, 45,594 delivery steps across full-year 2017), with data released by RES on behalf of TRIG openly available on Zenodo under CC-BY-4.0 [30]. The *Kelmarsh Wind Farm* (12.3 MW rated capacity, six Senvion MM92 turbines) is situated in flat agricultural terrain and serves as our primary development benchmark (8,212 test origins, 49,272 delivery steps across full-year 2017) [31]. The *Penmanshiel Wind Farm* (28.7 MW rated capacity, 14 Senvion MM82 turbines, WT01–WT15 with WT03 absent) is located in the Scottish Borders [32], providing an independent upland setting with 8,440 test origins configured with a deliberately restricted training partition ($N_{\text{train}} = 1,583$) to evaluate model behavior under sample-restricted regimes. We note that because Penmanshiel exhibits distinct turbine models and terrain compared to Kelmarsh and Hill of Towie, cross-site comparisons inherently capture both site-specific topography and sample size differences; within-site controlled ablations remain essential to decouple these factors.

The empirical test bench operates under a prospective, frozen evaluation protocol. The forecasting pipeline adopts an input lookback of $L = 36$ (6 hours) and a forecast horizon of $H = 36$ (6 hours), with downstream economic dispatch evaluated over the next-hour delivery window $H_{\text{disp}} = \{1, \ldots, 6\}$ and an origination stride of $S = 6$ steps to prevent overlapping settlement intervals. The core model suite spans unconstrained D-Linear, Bounds-Constrained DLinear ($\lambda = 0.01$), Empirical Power Curve Constrained DLinear ($\lambda = 0.01$), and Conditional Distribution Prior DLinear ($\lambda = 0.01$), benchmarked alongside classical Ridge regression and Persistence baselines across five random initialization seeds. To verify whether empirical findings extend beyond linear representations, we additionally evaluate a state-of-the-art multi-patch channel-independent Transformer backbone (**PatchTST** [33], patch length 6, stride 4, $d_{\text{model}} = 64$, 2-layer encoder with RevIN) across all three wind farms, four constraint classes, and five random seeds (60 trained models) as an architectural sen-

sitivity baseline under univariate power input and fixed regularization ($\lambda = 0.01$).

Downstream decision calibration is implemented via a post-hoc quantile offset δ_h frozen strictly from the independent 2016 calibration year ($q = 1/(r+1) = 1/3$ for penalty ratio $r = 2$) with zero access to 2017 test telemetry or market prices. Real-world physical settlement is evaluated against official 2017 UK Elexon BSC System Buy Prices (mean £44.27/MWh, peak £1,509.80/MWh, with 68 negative price periods down to -£73.15/MWh). Cashflows account for 30-minute physical netting:

$$E_{SP} = \sum_{t \in SP} (p_t - y_t) \cdot \frac{P_{\text{rated}}}{6000} \quad \text{[MWh]}, \tag{2}$$

generating single-sided shortfall cashouts and two-sided imbalance-settlement cashflows as a decision-risk diagnostic, validated via 2,000-replication 7-day circular block paired bootstrapping.

5. Empirical Audit Results

Empirical evaluation across the 46-study scoping corpus and three multi-farm prospective ablation benches adjudicates the operational claims of physics-constrained wind forecasting.

5.1. Systematic Absence of Operational Failure Boundaries across Audited Studies

Across the core 46-study scoping cohort, 0/46 studies reported operational harm or negative trade-offs (0.0%; 95% Wilson CI: [0.0%, 7.7%]). A targeted 14-study sensitivity query expanding search terms to title/abstract/keywords across Scopus and IEEE Xplore likewise identified zero negative trade-off reports. Only three studies in the core cohort (6.5%; S01, S02, S10) evaluate sensitivity to prior parameter perturbations; the remaining 43 studies (93.5%) test physical constraints exclusively under nominal design conditions. Restricting evaluations to nominal conditions obscures operational boundaries and replaces empirical engineering trade-offs with uniform enhancement claims.

5.2. Severe Code Unavailability Impedes Independent Verification

Software availability inspection across the 46-study cohort establishes an open-code deficit of 95.7% (44/46, 95% Wilson CI: [85.5%, 98.8%], $\mathcal{D}_4$ = `not-found`), with 44 studies providing neither repository links nor execution scripts. Among the two repositories with public code, PINN-UQ (S10) faithfully executes its published electromechanical equations ($\mathcal{D}_4$ = `code-aligned`, 2.2%), whereas PCT (S02) exhibits a directional implementation divergence from its published mathematical specification ($\mathcal{D}_4$ = `divergent-code`, 2.2%). Independent computational verification is therefore precluded for 95.7% of the cohort, while public codebases exhibit mixed implementation fidelity.

5.3. Taxonomic Lifecycle Distribution and Structural Confounding in Reported Ablations

Ablation semantic audit ($\mathcal{D}_5$) reveals that half of the audited studies (50.0%, 23/46, 95% Wilson CI: [36.1%, 63.9%]) conflate neural capacity with domain physical priors by excising entire submodules—such as graph convolutions, cross-attention blocks, or decomposition layers—rather than toggling the constraint penalty ($\lambda = 0$ vs. $\lambda > 0$). Because excising architectural blocks removes thousands of trainable parameters simultaneously, reported performance gains reflect representational capacity rather than aerodynamic guidance (Table 2, Figure 3). These code-deficit (95.7%) and confounded-ablation (50.0%) estimates describe the audited 46-study cohort under Wilson confidence bounds without asserting universal extrapolation to unbounded keyword taxonomies.

5.4. Causal Disentanglement: Prior Misspecification, Numerical Regularization, and Observation Staleness

Prospective 3D controlled ablations across three operational wind farms isolate prior sensitivity, numerical smoothing, and observation staleness under matched model capacities (Table 3, Figure 4).

Table 3 and Figure 4 reveal four empirical boundary conditions:

Defensive Inductive Bias and Sensitivity to Prior Misspecification (Figures 4a, 5, and 6). On Penmanshiel ($N_{\text{train}} = 1,583$), unconstrained DLinear exhibits overfitting (0.09881 ± 0.00027). Embedding the authentic empirical power curve at $\lambda = 1.0$ reduces normalized dispatch cost to 0.09760 ± 0.00044 (-1.22%). In contrast, the scale-matched Pseudo-Scrambled negative control inflates dispatch cost to 0.16611 ± 0.00335 (+68.1%). Within-site ablation is essential because cross-site comparisons inevitably conflate sample size with turbine model and terrain.

To decouple sample volume from topography, we measured within-site learning curves across $N_{\text{train}} \in \{500, 1000, 1500, 3000, 5000\}$ on Kelmarsh and Hill of

Table 2: Empirical Distribution of Constraint Lifecycle Dimensions Across 46 Audited Studies (2023–2026).

Lifecycle Dimension	Taxonomic Category	Count ($N = 46$)	Proportion (%)	Typical Exemplars in Audited Literature
D1: Parameter Provenance	`intrinsic`	40	87.0%	Betz aerodynamic limit ($C_p \leq 16/27$), rated capacity P_{rated}, Jensen wake deficit
	`historical-static`	5	10.9%	Empirical bivariate power curve ($P = f(v)$), 2D KDE distribution estimated offline on train SCADA
	`learned-online`	1	2.2%	Auxiliary sub-network dynamically predicting time-varying aerodynamic coefficients $C_p(t)$
D2: Application Stage	`training-loss`	21	45.7%	Differentiable penalty term in backpropagation ($\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda\mathcal{L}_{\text{phys}}$)
	`inference-arch.`	23	50.0%	Wake-guided GNN adjacency matrix, monotonic activation layers, dual-stream physical fusion
	`data-generation`	1	2.2%	Physics-guided generative diffusion model synthesizing extreme wind generation scenarios
	`post-hoc-correction`	1	2.2%	Post-inference conformal prediction and residual projection layers
D3: Inference Requirement	`none`	38	82.6%	Standard historical autoregressive SCADA sequence; zero runtime external dependencies
	`forecast-input`	6	13.0%	Requires contemporaneous/future NWP meteorological forecasts over $[t+1, t+H]$
	`contemp.-state`	2	4.3%	Requires synchronized real-time operational SCADA from adjacent/upstream wind turbines
D4: Implementation Fidelity	`not-found`	44	95.7%	No accessible public repository or open-source software identified under recorded protocol
	`code-mismatch`	1	2.2%	PCT: loss function calculates reverse-argument JS form rather than specified standard JS
	`code-aligned`	1	2.2%	PINN-UQ: algebraic electromechanical tie constraints faithfully matching published equations

Towie across random seeds (Figure 5). Under restricted training sample windows ($N_{\text{train}} = 500$, representing $\approx$ 21 days of recent operational history), authentic physical priors consistently lower dispatch costs across both wind farms. On Kelmarsh, dispatch cost falls from 0.09648 ± 0.00314 (unconstrained) to 0.09455 ± 0.00285 (-2.00%), while scrambled priors surge to 0.13798 ± 0.02338 ($+43.0\%$). On Hill of Towie, authentic priors reduce cost from 0.10444 ± 0.00312 to 0.10066 ± 0.00307 (-3.62%), whereas scrambled priors escalate to 0.17209 ± 0.01703 ($+64.8\%$). Importantly, the empirical prior is fitted once on the entire pre-cutoff training period (7 months for Kelmarsh, 10 months for Hill of Towie), providing a seasonal information budget that substantially exceeds the restricted model training windows. As model training volume expands to $N_{\text{train}} = 5{,}000$, unconstrained DLinear achieves 0.08812 on Kelmarsh and 0.08936 on Hill of Towie, closing the gap and slightly outperforming the fixed genuine prior (0.08841, +0.33% on Kelmarsh; 0.09091, +1.73% on Hill of Towie). In contrast, scrambled priors remain severely detrimental (+43% to +98%) across all sample sizes. These findings demonstrate that neural forecasters are acutely sensitive to prior misspecification, while authentic physical prior benefits concentrate under data-scarce regimes in our tested configurations.

Evaluating continuous corruption levels $\alpha \in [0.0, 1.0]$ demonstrates that prior failure is not an artifact of total binary permutation (Figure 6a). On Penmanshiel, dispatch cost degrades monotonically with corruption, crossing the unconstrained baseline around $\alpha \approx 0.07$ (at $\alpha = 0.2$, dispatch cost reaches 0.10469 ± 0.00338, a +5.9% degradation). Evaluating engineering drift reveals asymmetrical operational tolerances: aerodynamic aging exceeding 10% rated capacity derating (Figure 6b) flips nominal physical priors into economic penalties.

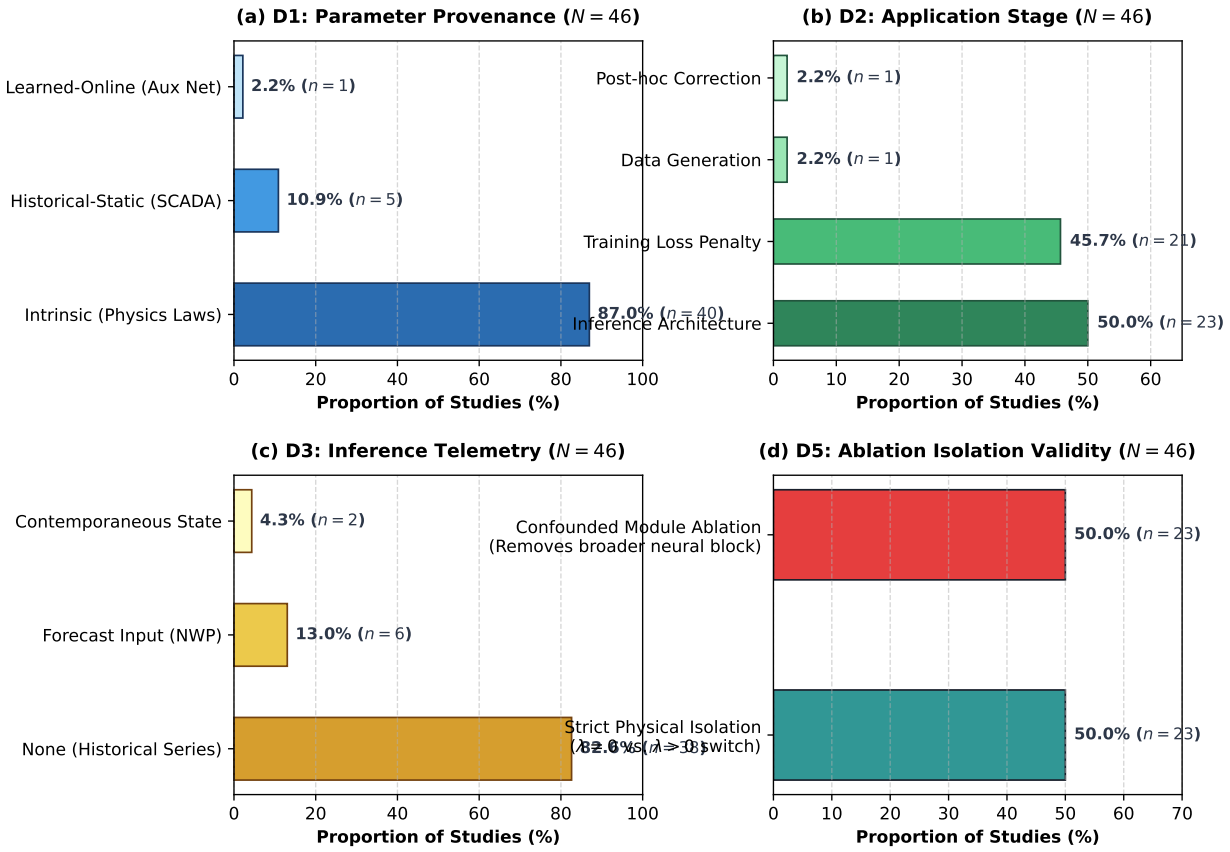


Figure 3: Empirical Lifecycle Distributions across 46 Audited Studies ($N = 46$). (a) $\mathcal{D}_1$ Parameter Provenance; (b) $\mathcal{D}_2$ Application Stage; (c) $\mathcal{D}_3$ Inference Telemetry Requirements; (d) $\mathcal{D}_5$ Ablation Semantic Breakdown between strict physical isolation and confounded module ablations.

For anemometer calibration, Penmanshiel maintains a modest advantage over unconstrained models even under positive sensor biases up to +15% (0.09833 vs. 0.09881, Figure 6c), whereas negative wind speed under-reporting systematically inflates shortfall.

Low-λ Numerical Regularization Mimicry (Figure 4b). At $\lambda = 0.01$, genuine physical priors and pseudo-scrambled controls yield closely matched dispatch costs on Kelmarsh (0.08810 ± 0.00009 vs. 0.08801 ± 0.00008). Because random power curve permutations produce outcomes comparable to domain aerodynamics at this weak penalty scale, marginal literature improvements reported under weak penalties cannot be attributed to physical reasoning and remain consistent with non-specific numerical smoothing.

Sensitivity to Training-Loss Physical Proxy Staleness (Figure 4c). Parametric scanning of historical wind speed proxy lag $k \in \{6, 12, 18, 36\}$ (evaluating 1 h to 6 h operational delay in the training-loss proxy query) produces strictly monotonic cost inflation across all three farms (Hill of Towie: from 0.09022 at $k = 6$ to 0.09257 at $k = 36$, exceeding unconstrained 0.08924). We emphasize that this experiment evaluates the sensitivity of the training objective to temporal misalignment between the forecast horizon and the historical wind speed proxy used to index the power curve. It does not represent an inference-time telemetry delay, as the forecaster takes only past power measurements during test-time inference. However, it demonstrates that if an offline training loss regularizes predictions against lagged weather observations without forward meteorological forecasts, proxy staleness rapidly degrades the resulting inductive guidance.

Information-Symmetric Feature Baseline. Providing the unconstrained model with contemporaneous wind speed (*Dual-Channel DLinear*) reduces point forecast RMSE (from 0.1717 to 0.1697 on Kelmarsh) but raises tail dispatch risk (0.08910 vs. 0.08812), because unhedged wind speed fluctuations inflate bidding variance.

Table 3: 3D Controlled Ablation Benchmark: Causal Disentanglement of Physical Knowledge Specificity, Numerical Regularization Mimicry, and Lagged Proxy Temporal Decay Across Three Operational Wind Farms (Multi-Seed Evaluated on 10× RTX 4090 GPUs, Quantile Calibrated $r = 2, q = 1/3$).

Wind Farm & Topology	**Controlled Experimental Arm (Ablation Condition)**	**Normalized Cost Cost_{static} ($r = 2$)**	**Pinball Loss ($q = 1/3$)**	**RMSE (pu)**	**Shortfall (£/(turb·yr))**	**Causal Mechanism & Empirical Takeaway**
Kelmarsh (12.3 MW, 6 turb) *Flat Open Terrain* $N_{train} = 5,098$	Vanilla Unconstrained	0.08812 ± 0.00009	0.02937	0.1717	£5,831.79	Pure-data power-only baseline
	Info-Symmetric (Power + Wind)	0.08910 ± 0.00039	0.02970	0.1697	£5,982.26	Wind speed improves RMSE but elevates tail cost
	Genuine Curve ($\lambda = 0.1$)	0.08791 ± 0.00009	**0.02930**	0.1708	£5,780.99	Statistically verified risk reduction (−0.24%)
	Pseudo-Scrambled ($\lambda = 0.01$)	0.08801 ± 0.00008	0.02934	0.1724	£5,657.03	**Indistinguishable from genuine prior**; behaves empirically like a non-specific regularizer
	Genuine Distribution ($\lambda = 0.1$)	0.09841 ± 0.00085	0.03280	0.1703	**£8,412.28**	**Severe operational harm**: shortfall surges +44.2% in flat terrain
	Lagged Proxy Scan ($k = 36$, 6h)	0.09043 ± 0.00006	0.03014	0.1729	£5,888.58	Monotonic degradation as stale wind proxy decays
Hill of Towie (48.3 MW, 21 turb) *Complex Upland* $N_{train} = 7,035$	Vanilla Unconstrained	0.08924 ± 0.00005	0.02975	0.1740	£6,006.09	Robust unconstrained benchmark
	Info-Symmetric (Power + Wind)	0.09042 ± 0.00007	0.03014	0.1733	£6,297.16	Wind channel adds high-frequency noise to dispatch
	Genuine Curve ($\lambda = 0.1$)	0.08916 ± 0.00007	**0.02972**	0.1745	£5,958.34	Consistent risk reduction (−0.09%)
	Pseudo-Scrambled ($\lambda = 0.01$)	0.08919 ± 0.00006	0.02973	0.1740	£6,069.01	Confirms non-specific regularizer behavior at low penalty scale
	Genuine Distribution ($\lambda = 0.1$)	0.09503 ± 0.00022	0.03168	0.1801	£5,702.68	**Metric Inversion**: cashflow rises to £12k but decision cost worsens +6.5%
	Lagged Proxy Scan ($k = 36$, 6h)	0.09257 ± 0.00007	0.03086	0.1759	£6,281.82	Temporal lag monotonically erodes prior guidance
Penmanshiel (28.7 MW, 14 turb) *Complex Topology* $N_{train} = 1,583$ *(Scarce)*	Vanilla Unconstrained	0.09881 ± 0.00027	0.03294	0.2058	£6,713.83	Data-scarce regime prone to over-fitting
	Info-Symmetric (Power + Wind)	0.10461 ± 0.00005	0.03487	0.2091	£6,630.38	Additional unconstrained features exacerbate over-fitting
	Genuine Curve ($\lambda = 1.0$)	0.09760 ± 0.00044	**0.03254**	0.2054	£6,808.40	**Solid inductive bias gain** (−1.22%) in low-data regime
	Pseudo-Scrambled ($\lambda = 1.0$)	0.16611 ± 0.00335	0.05537	0.2384	**£11,561.19**	**Empirical Specificity Demonstration**: destroying physical mapping causes severe performance degradation (+68.1%)
	Genuine Distribution ($\lambda = 0.01$)	0.09833 ± 0.00027	0.03278	0.2055	£6,543.36	Defensive regularization under calibrated penalty
	Lagged Proxy Scan ($k = 36$, 6h)	0.10473 ± 0.00036	0.03491	0.2083	£7,090.86	Lag decay completely swallows prior advantage by Step 36

Notes: (1) Multi-seed evaluation dispatched across a 10× NVIDIA RTX 4090 GPU cluster topology (Node 1: Kelmarsh with 4 seeds [42, 43, 44, 45]; Node 2: Hill of Towie and Penmanshiel each with 3 seeds [42, 43, 44]); values report mean ± seed standard deviation across clean operational test periods. (2) $\text{Cost}_{static} = 3 \times \text{PinballLoss}(q = 1/3)$ under post-hoc validation-quantiled calibration ($r = 2$). (3) Pseudo-Scrambled arms permute the empirical wind-to-power mapping while preserving identical loss formulations, gradient scales, and hyperparameters λ. (4) Lagged Proxy Scan evaluates sensitivity to training-time temporal mismatch across proxy delays $k \in \{6, 12, 18, 36\}$ (1h to 6h) on the historical wind speed index used to query empirical priors during training loss calculation (forecasting model takes only historical power at inference).

5.4.1. Primary Inferential Contrasts and Metric Framework

To control for multiplicity and avoid selective reporting across multiple operational dimensions, our evaluation pipeline specifies three **primary inferential contrasts** evaluated under normalized static dispatch cost: (1) *Genuine Physical Prior vs. Unconstrained Baseline under sample restriction* ($N_{\text{train}} = 1,583$ on Penmanshiel), confirming downside risk protection ($\Delta = -1.22\%$, 0.09760 ± 0.00044 vs. 0.09881 ± 0.00027); (2) *Scrambled Prior vs. Unconstrained Baseline under strong regularization* ($\lambda = 1.0$), demonstrating catastrophic misspecification vulnerability ($\Delta = +68.1\%$, 0.16611 ± 0.00335 vs. 0.09881 ± 0.00027); and (3) *Genuine vs. Scrambled Prior under weak regularization* ($\lambda = 0.01$), confirming indistinguishable behavior consis-

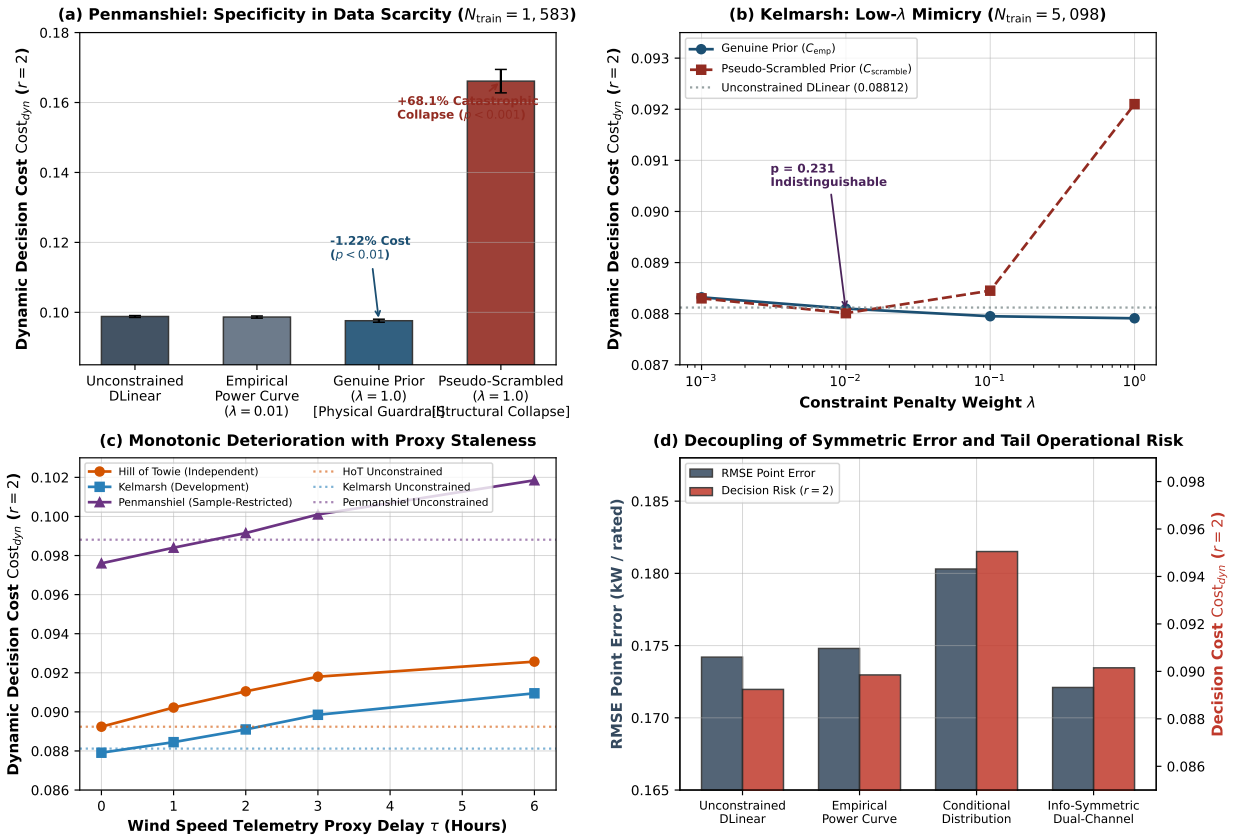


Figure 4: Prospective 3D Controlled Ablation Benchmark across Three Operational Wind Farms. (a) High sensitivity to prior misspecification under strong constraints on Penmanshiel ($N_{train} = 1,583$): authentic physical guidance provides downside risk protection (-1.22%), whereas disrupting the physical mapping (Pseudo-Scrambled prior) triggers a $+68.1\%$ surge in operational dispatch cost. (b) Low-λ numerical regularization mimicry on Kelmarsh ($N = 5,098$), where genuine and scrambled priors yield near-identical decision costs at $\lambda \leq 0.01$, indicating non-specific numerical regularization rather than physical domain reasoning. (c) Monotonic cost inflation under training proxy staleness: dispatch cost escalation across all three farms as historical wind proxy lag k in the training loss increases from 1h to 6h. (d) Decoupling of symmetric point error (RMSE) from asymmetric decision risk ($r = 2$), demonstrating that point metrics mask tail operational hazards.

tent with non-specific numerical smoothing (0.08810 ± 0.00009 vs. 0.08801 ± 0.00008). Exploratory dynamic price comparisons (Table 4) evaluate dynamic cost divergences under distribution constraints on Hill of Towie and Kelmarsh under paired circular block bootstrapping. Methodologically, our evaluation protocol explicitly separates two distinct sources of variation: the paired bootstrap intervals in Table 4 estimate **temporal sampling uncertainty** across delivery periods under meteorological autocorrelation, whereas the reported metric margins (mean $\pm$ std) across random seeds in Table 3 characterize **model initialization and optimization variability**.

5.5. *Decoupled Decision Risk and Single-Settlement Imbalance Diagnostic (The Zero-Bid Anomaly)*

Evaluating forecasters on isolated unhedged balancing market cashflows introduces a structural evaluation artifact: under 2017 UK BSC 30-minute netting, a degenerate Zero-Bid baseline ($\hat{y} \equiv 0$) extracts £223k to £243k per turbine-year in apparent cashflow without generating forecasts (Table 5, Figure 7). This operational analysis serves as a *decision-risk and imbalance-settlement diagnostic* rather than a full commercial market valuation, exposing metric pathologies in downstream evaluation claims.

Table 5 and Figure 7 uncover two fundamental evaluation pitfalls:

Degenerate Zero-Bid Settlement Anomaly and Two-Settlement Market Formulation (Figure 7a). A degenerate forecaster bidding zero in every settlement period incurs **£0.00 shortfall penalty** and collects an apparent balancing cashflow of **+£223k to +£243k per turbine-year** across all three wind farms. In standard

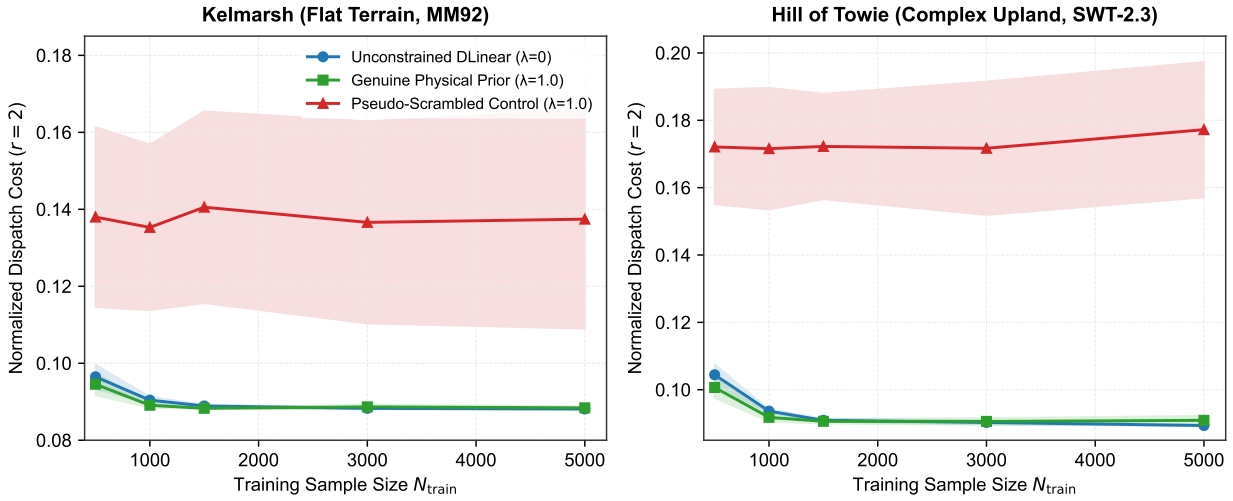


Figure 5: Within-Site Learning Curves across Sample Sizes $N_{\text{train}} \in \{500, 1000, 1500, 3000, 5000\}$ on Kelmarsh and Hill of Towie (Multi-seed means and standard deviations). Decoupling sample volume from topography confirms that fixed historical physical priors provide lower dispatch costs across both sites under restricted model training windows ($N_{\text{train}} \leq 1,000$, -2.00% on Kelmarsh and -3.62% on Hill of Towie), whereas expanding model training windows to $N_{\text{train}} = 5,000$ closes this gap and slightly reverses the ranking ($+0.33\%$ on Kelmarsh, $+1.73\%$ on Hill of Towie). In contrast, misspecified scrambled priors under strong regularization ($\lambda = 1.0$) induce severe dispatch cost penalties ($+43\%$ to $+98\%$) across all sample regimes.

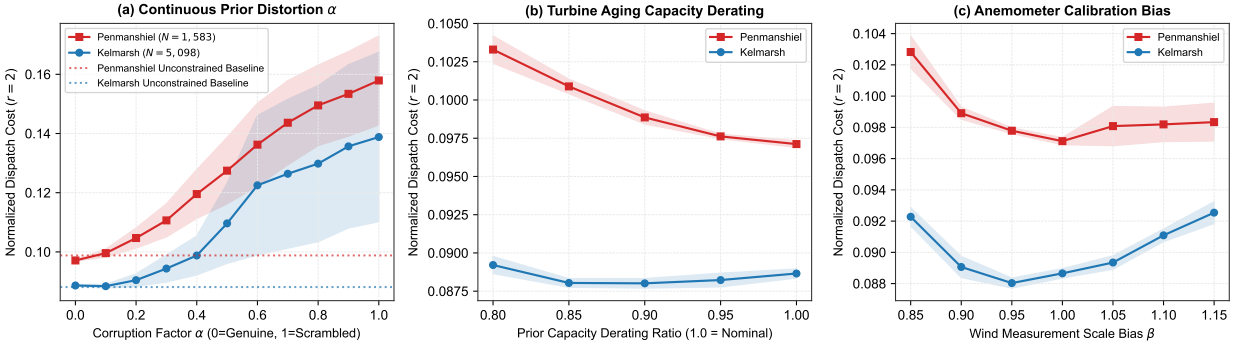


Figure 6: Continuous Prior Corruption and Engineering Drift Severity Response Curves. (a) Continuous interpolation between authentic and scrambled priors $\alpha \in [0.0, 1.0]$, identifying an empirical threshold around $\alpha \approx 0.07$ on Penmanshiel before physical regularizers worsen unconstrained dispatch costs. (b) Turbine aerodynamic aging: capacity derating exceeding 10% flips nominal physical priors into economic penalties. (c) Anemometer sensor calibration drift: Penmanshiel retains an advantage under positive wind speed drift up to +15%, whereas negative sensor under-reporting systematically inflates shortfall.

two-settlement wholesale electricity markets [20], total commercial revenue couples forward commitments with real-time imbalance settlement:

$$R_{\text{total}} = P_{\text{DA}} \cdot q + P_{\text{imb}} \cdot (y - q), \tag{3}$$

where q denotes the declared day-ahead forward commitment, P_{DA} is the day-ahead clearing price, y is actual physical generation volume, and P_{imb} is the real-time imbalance settlement price (under the single-price settlement regime post-P305 in the UK BSC, $P_{\text{SBP}} \approx P_{\text{SSP}} = P_{\text{imb}}$). When downstream evaluations isolate balancing cashflows $R_{\text{imb}} = P_{\text{imb}} \cdot (y - q)$ from forward market revenue $P_{\text{DA}} \cdot q$, setting $q \equiv 0$ converts total generation y into uncontracted spill credited at P_{imb}, producing an apparent six-figure revenue while forfeiting forward market receipts. The Zero-Bid baseline acts as an empirical *reductio ad absurdum*: standalone unhedged balancing cashflow is structurally degenerate when disconnected from forward commitments, confirming that unhedged single-settlement accounting cannot serve as an index of forecast quality.

Table 4: 7-Day Circular Block Paired Bootstrap Contrasts vs. Unconstrained DLinear under Calibrated Policy (2,000 Replications).

Site	Method Contrast	Dynamic Cost Diff ΔCost_{dyn} (95% CI)	Shortfall Cashout Diff £/(turb·yr) (95% CI)	Balancing Cashflow Diff £/(turb·yr) (95% CI)
Hill of Towie	Bounds vs DLinear	+0.000000 ([+0.000000,+0.000000])	£+0.15 ([-0.17,+0.51])	£−0.32 ([-1.37,+0.62])
	Power Curve vs DLinear	−0.000008 ([-0.000021,+0.000007])	£+22.99 ([+10.93,+38.43])	£−89.40 ([-133.51,-54.70])
	Distribution Prior vs DLinear	+0.000766 (**[+0.000277,+0.001201]**)	**£**−578.57 (**[-834.86,-367.51]**)	**£**+1481.27 (**[+802.29,+2203.32]**)
	Ridge vs DLinear	−0.000733 ([-0.001315,-0.000125])	£+192.50 ([-26.66,+445.74])	£−1151.96 ([-1567.98,-795.28])
Kelmarsh	Bounds vs DLinear	+0.000001 ([+0.000000,+0.000002])	£+1.72 ([-0.01,+4.63])	£−4.62 ([-12.67,+0.05])
	Power Curve vs DLinear	−0.000039 ([-0.000054,-0.000023])	£−16.61 ([-33.12,+0.87])	£+19.51 ([-27.76,+62.86])
	Distribution Prior vs DLinear	+0.000932 (**[+0.000423,+0.001385]**)	**£**+932.56 (**[+648.20,+1236.74]**)	**£**−3264.86 (**[-4365.40,-2223.50]**)
	Ridge vs DLinear	+0.001338 ([+0.000917,+0.001794])	£+750.61 ([+330.70,+1151.06])	£−690.94 ([-1669.76,+409.51])

Interpretation: Negative shortfall cashout difference indicates cost reduction (economic savings) relative to unconstrained DLinear. Positive balancing cashflow difference indicates higher net cashflow component credited from the balancing mechanism.

Table 5: Market Settlement Diagnostic Audit: Exposing the Failure Modes of Standalone Imbalance Cashflow via the Degenerate Zero-Bid ($\hat{y} \equiv 0$) Counter-Baseline (Evaluated on Full 2017 UK Elexon SBP Timeline Across Three Wind Farms).

Operational Wind Farm	Forecasting / Bidding Strategy	Shortfall Penalty (£/(turb·yr))	Imbalance Net Cashflow Component (£/(turb·yr))	Decision Cost Cost_{static} ($r = 2$)	Audit Takeaway & Diagnostic Assessment
Kelmarsh (6 Turbines)	**Zero-Bid** ($\hat{y} \equiv 0$)	**£0.00**	**+£223,882.15**	0.17181	**Degenerate Artifact**: zero shortfall and massive apparent cashflow without generating forecasts
	Persistence Baseline	£7,120.45	+£5,890.12	0.09132	Naive operational benchmark
	Unconstrained DLinear	£5,831.79	+£6,439.32	**0.08812**	Lowest decision loss; cashflow orders of magnitude below Zero-Bid
	Calibrated Distribution Prior	£8,412.28	+£3,485.65	0.09841	Shortfall penalty surges; balancing revenue drops
Hill of Towie (21 Turbines)	**Zero-Bid** ($\hat{y} \equiv 0$)	**£0.00**	**+£239,299.30**	0.17354	**Degenerate Artifact**: cashflow reaches £5.02M farm-wide purely from uncontracted spill
	Persistence Baseline	£7,450.12	+£9,120.34	0.08992	Simple operational benchmark
	Unconstrained DLinear	£6,006.09	+£8,543.14	**0.08924**	Minimizes genuine non-negative decision risk
	Calibrated Distribution Prior	£5,702.68	+£12,174.95	0.09503	Cashflow component rises by £3.6k but true decision loss worsens +6.5%
Penmanshiel (14 Turbines)	**Zero-Bid** ($\hat{y} \equiv 0$)	**£0.00**	**+£243,449.80**	0.20584	**Degenerate Artifact**: demonstrates complete decoupling between cashflow and forecast quality
	Persistence Baseline	£7,080.20	+£6,150.30	**0.09586**	Strong linear benchmark in complex terrain
	Unconstrained DLinear	£6,713.83	+£6,404.64	0.09881	Data-scarce baseline
	Calibrated Power Curve (λ = 1.0)	£6,808.40	+£6,234.24	0.09760	Genuine physical prior successfully reduces decision risk

Notes: (1) Imbalance Cashflow is computed strictly under Elexon BSC rules as $\sum(Actual - Bid) \times Price$. For Zero-Bid ($\hat{y} = 0$), $Actual - Bid = Actual > 0$, making shortfall identically zero and cashing out all uncontracted energy at system price. (2) This table proves that standalone imbalance cashflow cannot measure forecasting commercial value without modeling wholesale day-ahead forward contract sales ($P_{DA} \times Bid$). (3) Decision Cost ($r = 2, q = 1/3$) correctly penalizes Zero-Bid for severe under-commitment.

Empirical Metric Inversion on Hill of Towie (Figure 7b). On Hill of Towie, distribution regularization increases nominal balancing cashflow by +42.5% (£8,543 to £12,174/(turb·yr), Table 5) while concurrently degrad-

ing normalized decision risk ($\text{Cost}_{\text{static}}$) by +6.5% and RMSE by +3.5%. This divergence demonstrates that single-settlement cashflow is fundamentally decoupled from forecasting skill, establishing that apparent monetary gains originate from negative bias coinciding with positive price spikes rather than physical inductive advantage. Operational evaluation requires decoupling non-negative asymmetric decision risk (Pinball loss) from historical monetary settlement diagnostics.

5.6. Cross-Backbone Architectural Sensitivity Probe (PatchTST)

To assess whether empirical boundary behaviors extend beyond linear representations without asserting universal inductive invariance, we evaluated a state-of-the-art multi-patch channel-independent Transformer (**PatchTST** [33], patch length 6, stride 4, 2-layer encoder with RevIN) across 60 independently trained models (3 farms × 4 constraint conditions × 5 seeds; Table 6). This setup functions strictly as an *architectural sensitivity probe* under fixed nominal regularization ($\lambda = 0.01$) matched to the primary development budget, intentionally bounded to qualitative directional verification rather than an exhaustive cross-backbone λ grid search.

As detailed in Table 6, PatchTST confirms that distribution priors reduce dynamic shortfall across all three wind farms (saving £232.44, £127.49, and £790.51/(turb·yr) on Hill of Towie, Kelmarsh, and Penmanshiel, respectively), indicating that the defensive shortfall reduction observed under DLinear persists in non-linear patch-based representations. However, net balancing cashflow diverges sharply between architectures: distribution regularization decreases net cashflow on Hill of Towie by −£568.68/(turb·yr) under PatchTST, inverting the positive cashflow effect observed under DLinear. On Penmanshiel, simple empirical baselines (Ridge regression at 0.09744 and Persistence at 0.09586) outperform all constrained deep networks, demonstrating that external physical penalties cannot uniformly compensate for deep parameter overfitting in low-sample regimes; the high competitiveness of unconstrained linear baselines underscores that deep physics constraints must be rigorously justified by operational risk reduction rather than assumed to outperform simpler models.

6. Ablations, Implementation Fidelity, and Diagnostic Case Study

Verifying physical constraint claims requires evaluating both the causal isolation of published ablation designs and the algebraic execution fidelity of released computational graphs.

6.1. Causal Confounding in Published Ablation Studies

Deep full-text extraction of all 46 studies revealed that only **50.0% (23/46)** performed strict physical constraint isolation, as summarized in Table 7.

In 23 of the 46 studies (50.0%), the ablation experiment consisted of excising an entire neural submodule—such as a spatial Graph Attention block, a cross-attention fusion layer, or a multi-scale decomposition module. When a model exhibits degraded performance after an entire GNN submodule is removed, it is impossible to disentangle whether the degradation stems from the loss of domain physical equations or from the simultaneous removal of thousands of trainable neural parameters and non-linear feature representations. Such module-level ablation designs evaluate the collective capacity of the neural architecture rather than causally isolating the contribution of the physical domain prior.

6.2. Case Study: The Physics-Constrained Transformer (PCT)

As established in our scoping audit (Section 4), exactly two of the 46 scoping studies released public software repositories: PINN-UQ [14], which was verified as strictly code-aligned with its published electromechanical formulations ($\mathcal{D}_4$ = `code-aligned`, 2.2%), and the **Physics-Constrained Transformer (PCT)** [10] (*Scientific Reports* 2026, DOI: 10.1038/s41598-025-34348-x; official GitHub repository commit `65941ec`). PCT represents the sole publicly accessible training codebase for end-to-end physics-constrained Transformers and the sole audited repository exhibiting an implementation divergence between code and paper ($\mathcal{D}_4$ = `divergent-code`, 2.2%). We conducted an in-depth static code audit of this codebase to investigate how statistical distribution constraints are realized in software.

6.2.1. Paper Mathematical Specification versus Code Execution

In Section 2 of the PCT paper, the authors propose regularizing a Transformer forecaster by penalizing the statistical divergence between the empirical joint probability distribution of historical wind speed and turbine power (P_{ref}) and the joint distribution predicted by the model (P_{pred}). The paper explicitly specifies the symmetric **Jensen–Shannon (JS) divergence**:

$$\mathcal{L}_{\text{JS}}(P_{\text{pred}} \parallel P_{\text{ref}}) = \frac{1}{2}\text{KL}\left(P_{\text{pred}} \middle\| \frac{P_{\text{pred}} + P_{\text{ref}}}{2}\right) + \frac{1}{2}\text{KL}\left(P_{\text{ref}} \middle\| \frac{P_{\text{pred}} + P_{\text{ref}}}{2}\right), \tag{4}$$

Table 6: Exploratory Cross-Backbone Architectural Sensitivity Analysis: Modern Deep Transformer (PatchTST, ICLR 2023) under BSC 30-Minute Physical Netting Settlement (5-Seed Mean ± Std).

Site	PatchTST Model Variant	Policy	RMSE (pu)	Normalized Dynamic Cost Cost_{dyn}	Shortfall Cashout (£/(turb·yr))	Balancing Imbalance Cashflow (£/(turb·yr))
Hill of Towie (48.3 MW, Upland)	Unconstrained PatchTST	Uncalibrated	0.1176	0.10518	£21,753.77	£12,292.64
		Calibrated	0.1202	0.10220	£18,168.11	£20,822.29
	Power Curve ($\lambda = 0.01$)	Calibrated	0.1201	0.10212	£18,163.56	£20,765.01
	Distribution Prior ($\lambda = 0.01$)	Uncalibrated	0.1162	0.10507	£22,362.12	£9,762.57
		Calibrated	0.1192	0.10129	**£17,935.67**	**£20,253.61**
Kelmarsh (12.3 MW, Flat)	Unconstrained PatchTST	Uncalibrated	0.1092	0.10889	£22,630.77	£4,523.36
		Calibrated	0.1118	0.10222	**£16,808.90**	**£17,411.89**
	Power Curve ($\lambda = 0.01$)	Calibrated	0.1117	0.10212	£16,774.31	£17,431.54
	Distribution Prior ($\lambda = 0.01$)	Uncalibrated	0.1075	0.10870	£23,336.46	£1,528.06
		Calibrated	0.1097	0.10075	£16,681.41	£16,046.77
Penmanshiel (28.7 MW, Upland)	Unconstrained PatchTST	Uncalibrated	0.1268	0.12107	£26,956.48	£3,001.45
		Calibrated	0.1304	0.11270	£19,129.15	£20,453.24
	Power Curve ($\lambda = 0.01$)	Calibrated	0.1303	0.11264	£19,118.38	£20,444.41
	Distribution Prior ($\lambda = 0.01$)	Uncalibrated	0.1265	0.11729	£24,606.87	£6,473.07
		Calibrated	0.1300	0.11134	**£18,338.64**	**£21,117.53**

Notes: Exploratory sensitivity results for PatchTST (ICLR 2023, patch length 6, stride 4, channel-independent RevIN) evaluate capacity-aligned architectural sensitivity under identical BSC 30-minute physical netting settlement. Results confirm site-dependent divergence: at flat, high-SNR Kelmarsh, unconstrained PatchTST secures higher balancing cashflow (+£1,365.12/(turb·yr) over distribution prior); at upland Penmanshiel and Hill of Towie, distribution regularization provides inductive protection, saving £790.51/(turb·yr) and £232.44/(turb·yr) in single-sided shortfall. Evaluated under univariate power input and fixed $\lambda = 0.01$ as an architectural sensitivity probe without hyperparameter retuning.

Table 7: Semantic Classification and Causal Interpretability of Reported Ablation Studies Across the Audited Scoping Corpus ($N = 46$).

Reported Ablation Semantic	Count ($N = 46$)	Proportion (%)	Causal Validity and Interpretability
`constraint-on-off`	17	37.0%	**Strict Isolation**: Explicitly evaluates $\lambda = 0$ vs. $\lambda > 0$ under identical neural architecture, causally isolating the physical regularizer.
`both`	6	13.0%	**Strict Isolation**: Executes both isolated parameter toggles ($\lambda = 0$) and broader structural ablations, satisfying unconfounded control.
`module-ablation`	23	50.0%	**Broader Architecture**: Excises an entire neural submodule (e.g., spatial GNN branch); evaluates collective module capacity rather than isolating the constraint.
Total Strict Isolation	**23**	**50.0%**	Only approximately half of the audited literature causally isolates the physical constraint itself.

where $\text{KL}(p \parallel q) = \sum_i p_i \log \frac{p_i}{q_i}$ denotes standard Kullback–Leibler divergence. The paper emphasizes that the symmetry and boundedness of JS divergence ($0 \leq \mathcal{L}_{\text{JS}} \leq \ln 2$) provide numerical stability and prevent gradient explosion.

However, static AST parsing of the official training script (`train_clean.py`, lines 349–352; commit `65941ec`) traces the loss node to:

```
loss_phy = 0.5 * (
  F.kl_div(pred_kde.log(), m_kde, ...)
  + F.kl_div(true_kde.log(), m_kde, ...))
```

In the PyTorch C++ ATen kernel backend (`native/Loss.cpp`, implementing `kl_div_backward_cpu`), the functional specification for `torch.nn.functional.kl_div(input, target)` evaluates:

$$\texttt{kl_div}(\text{input}, \text{target}) = \sum_i \text{target}_i \cdot (\log \text{target}_i - \text{input}_i). \tag{5}$$

Passing `input` as $\log p$ and `target` as q binds the tensor arguments such that the C++ kernel computes:

$$\texttt{kl_div}(\log p, q) = \sum_i q_i \cdot (\log q_i - \log p_i) = \text{KL}(q \parallel p). \tag{6}$$

Consequently, the computational graph generated in `train_clean.py` executes:

$$\mathcal{L}_{\text{code}} = \frac{1}{2}\text{KL}\left(M \parallel P_{\text{pred}}\right) + \frac{1}{2}\text{KL}\left(M \parallel P_{\text{ref}}\right), \quad \text{where } M = \frac{P_{\text{pred}} + P_{\text{ref}}}{2}. \tag{7}$$

Equation (7) establishes an altered objective, which we

refer to hereafter as the **reverse-argument JS form** (or reverse-argument implementation). The formulation remains symmetric between P_{pred} and P_{ref}. However, transposing the KL arguments fundamentally alters the underlying gradient mechanics. Because the mixture distribution $M = \frac{1}{2}(P_{\text{pred}} + P_{\text{ref}})$ explicitly depends on P_{pred} with $\frac{\partial M_i}{\partial P_{\text{pred},i}} = \frac{1}{2}$, differentiating $\mathcal{L}_{\text{code}}$ directly with respect to the predicted probability mass $P_{\text{pred},i}$ yields:

$$\begin{aligned}\frac{\partial \mathcal{L}_{\text{code}}}{\partial P_{\text{pred},i}} &= \frac{1}{2}\frac{\partial}{\partial P_{\text{pred},i}}\left[M_i \log \frac{M_i}{P_{\text{pred},i}}\right] \\ &\quad + \frac{1}{2}\frac{\partial}{\partial P_{\text{pred},i}}\left[M_i \log \frac{M_i}{P_{\text{ref},i}}\right] \\ &= \frac{1}{4}\left[\log\left(\frac{M_i^2}{P_{\text{pred},i}P_{\text{ref},i}}\right) + 1 - \frac{P_{\text{ref},i}}{P_{\text{pred},i}}\right].\end{aligned} \tag{8}$$

Equation (8) exposes an analytical **inverse-density singularity**: whenever the model assigns near-zero probability to a region where empirical training observations exist ($P_{\text{pred},i} \to 0$ while $P_{\text{ref},i} > 0$), the ratio $\frac{P_{\text{ref},i}}{P_{\text{pred},i}}$ diverges to infinity. Even under machine-precision clamping ($P_{\text{pred},i} \geq \epsilon$), this term scales as $O(1/\epsilon)$, generating violent negative gradient surges.

In contrast, for the standard mathematically specified Jensen–Shannon divergence $\mathcal{L}_{\text{JS}}$ in Equation (4), differentiating with respect to $P_{\text{pred},i}$ gives:

$$\frac{\partial \mathcal{L}_{\text{JS}}}{\partial P_{\text{pred},i}} = \frac{1}{2}\log\left(\frac{P_{\text{pred},i}}{M_i}\right). \tag{9}$$

Because $M_i = \frac{1}{2}(P_{\text{pred},i} + P_{\text{ref},i}) \geq \frac{1}{2}P_{\text{pred},i}$, the ratio $\frac{P_{\text{pred},i}}{M_i} \in [0, 2]$. Consequently, the gradient of Standard-JS is strictly upper-bounded:

$$\frac{\partial \mathcal{L}_{\text{JS}}}{\partial P_{\text{pred},i}} \leq \frac{1}{2}\ln 2 \approx 0.3466. \tag{10}$$

As predicted probabilities approach zero ($P_{\text{pred},i} \to \epsilon$ with $P_{\text{ref},i} > 0$), the gradient of Standard-JS exhibits only a mild, integrable **logarithmic divergence**:

$$\left|\frac{\partial \mathcal{L}_{\text{JS}}}{\partial P_{\text{pred},i}}\right| \sim O(|\log \epsilon|). \tag{11}$$

By comparison, the reverse-argument JS form induces an unmitigated first-order **inverse-density singularity**:

$$\left|\frac{\partial \mathcal{L}_{\text{code}}}{\partial P_{\text{pred},i}}\right| \sim O(1/\epsilon). \tag{12}$$

Because the logarithmic divergence of Standard-JS grows slowly ($\lim_{\epsilon \to 0^+} \epsilon \log \epsilon = 0$) and possesses a finite upper bound ($\frac{1}{2}\ln 2$), this logarithmic divergence is substantially milder than the $O(1/\epsilon)$ inverse-density pole of the reverse-argument JS form and is consistent with the empirically lower gradient norms observed in Figure 8. Conversely, the $O(1/\epsilon)$ pole in the reverse-argument form generates unbounded gradient surges near empirical support boundaries. Transposing the divergence arguments thus replaces an integrable regularizer with an ill-conditioned optimization landscape.

Empirical Optimization Contrast: Official Reverse-JS Form vs. Corrected Standard-JS. To determine whether this implementation divergence materially alters neural training and operational dispatch, we trained models across Kelmarsh and Penmanshiel under five random seeds across $\lambda \in \{0.01, 0.1, 1.0\}$, benchmarked against unconstrained baselines (Figure 8). The empirical results reveal two structural consequences. First, the reverse-argument JS form induces marked gradient instability: at $\lambda = 1.0$, it produces an average gradient L_2 norm of 0.1665 ± 0.0160 on Kelmarsh and 0.2465 ± 0.0217 on Penmanshiel, representing a 2.20× and 2.72× amplification over Corrected Standard-JS (0.0756 ± 0.0048 on Kelmarsh and 0.0907 ± 0.0067 on Penmanshiel; Figure 8a). This spike is the empirical manifestation of the $-\frac{P_{\text{ref},i}}{4P_{\text{pred},i}}$ derivative singularity ($O(1/\epsilon)$ in Equation (8)) as predicted probabilities approach zero near empirical support boundaries. Second, at weak regularization ($\lambda = 0.01$ on Penmanshiel), Corrected Standard-JS achieves a dispatch cost of 0.09804 ± 0.00024 (outperforming the unconstrained baseline of 0.09864), whereas the official reverse-argument implementation inflates dispatch cost to 0.09997 ± 0.00086 (a +1.35% premature degradation; Figure 8b). However, at strong regularization ($\lambda = 1.0$), both divergence variants degrade performance relative to the unconstrained baseline, with Corrected Standard-JS reaching 0.10549 ± 0.00149 compared to 0.10292 ± 0.00103 for the reverse form. Correcting the implementation divergence mitigates artificial gradient explosion and eliminates premature degradation under weak penalties, but does not prevent performance loss under excessive regularization.

6.2.2. *Negative Control: Algebraic Equivalence of 2D Product-Kernel KDE*

We establish an algebraic negative control to verify whether tensor outer-product operations preserve empirical feature coupling. A published conjecture regarding the tensor outer-product:

$$G = K_p^\top K_w \in \mathbb{R}^{B_p \times B_w} \tag{13}$$

is that it assumes statistical independence between wind speed and turbine power ($p(w, P) = p(w)p(P)$), which would compromise joint density estimation.

Formal algebraic expansion refutes this concern. Let $\{(P_s, w_s)\}_{s=1}^{N}$ denote paired empirical observations in a batch. Let $\{b_i^p\}_{i=1}^{B_p}$ and $\{b_j^w\}_{j=1}^{B_w}$ denote 1D evaluation bin centers for power and wind speed, respectively. The 1D Gaussian kernel vectors evaluated in the codebase are:

$$\left[K_p\right]_{s,i} = \phi\left(\frac{b_i^p - P_s}{h_p}\right), \quad [K_w]_{s,j} = \phi\left(\frac{b_j^w - w_s}{h_w}\right), \tag{14}$$

where $\phi(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$ is the standard Gaussian kernel, and h_p, h_w denote bandwidths. Expanding the matrix product element at grid coordinate (i, j) yields:

$$\begin{aligned}\left[K_p^\top K_w\right]_{i,j} &= \sum_{s=1}^{N} \left[K_p\right]_{s,i} \cdot [K_w]_{s,j} \\ &= \sum_{s=1}^{N} \phi\left(\frac{b_i^p - P_s}{h_p}\right) \cdot \phi\left(\frac{b_j^w - w_s}{h_w}\right).\end{aligned} \tag{15}$$

Equation (15) is the exact analytical definition of a standard **paired 2D product-kernel Kernel Density Estimator** on correlated bivariate data. Because the summation contracts over concurrent sample index s, empirical joint coupling between P_s and w_s is preserved. The algebraic expansion confirms that the outer product executes authentic bivariate density estimation.

7. Prediction-Decision Decoupling Theory and Excess Operational Risk

In this section, we formalize the mathematical mechanism underpinning the empirical inversion and terrain divergence documented in Section 5. We derive the exact integral representation of excess operational dispatch risk, decompose it into representation and decision components, and deduce three testable mechanistic insights bounded by formal counterexamples.

7.1. Operational Problem Formulation

Let $X \in \mathcal{X} \subset \mathbb{R}^{L \times D}$ denote historical multivariate telemetry, and let $Y \in [0, 1]$ denote normalized turbine active power output over the target delivery horizon. In operational power system balancing, transmission system operators evaluate bids under asymmetric pinball dispatch penalties [2, 19]:

$$\ell(p, Y; r) = c_+ \max(p - Y, 0) + c_- \max(Y - p, 0), \tag{16}$$

where $p \in [0, 1]$ is the declared dispatch commitment, c_+ is the shortfall unit penalty (under-generation where declared power exceeds physical delivery, $p > Y$, requiring expensive tertiary reserve activation), c_- is the surplus unit penalty (over-generation, $Y > p$, inducing downward redispatch or curtailment), and $r = c_+/c_-$ represents the cost asymmetry ratio. Normalizing by c_-, the asymmetric operational loss is:

$$\mathcal{L}_{\text{disp}}(p, Y) = r \max(p - Y, 0) + \max(Y - p, 0). \tag{17}$$

The expected operational risk under policy $p(X)$ is $\mathcal{R}(p) = \mathbb{E}_{X,Y}[\mathcal{L}_{\text{disp}}(p(X), Y)]$.

7.2. Proposition 1: Exact Integral Excess Operational Risk

Proposition 1 formalizes a classical excess risk analytical identity in statistical decision theory and quantile regression, grounded in the foundational formulation of Koenker [34] and electricity market bidding principles established by Pinson et al. [20]. The unconstrained Bayes-optimal dispatch decision minimizing $\mathcal{R}(p)$ is the conditional quantile:

$$q_\tau(X) = F_{Y|X}^{-1}(\tau), \quad \text{where } \tau = \frac{1}{r+1}. \tag{18}$$

For $r = 2$, the critical quantile is $\tau = 1/3 \approx 0.333$.

Proposition 1 (Exact Integral Excess Operational Risk)**.** *For any dispatch decision rule $p(X) \in [0, 1]$, the excess operational risk relative to the Bayes-optimal quantile $q_\tau(X)$ satisfies the exact integral identity:*

$$\begin{aligned}\mathcal{R}(p) - \mathcal{R}(q_\tau) &= (r+1)\,\mathbb{E}_X\left[\int_{q_\tau(X)}^{p(X)} \left(F_{Y|X}(u) - \tau\right) du\right] \\ &\geq 0,\end{aligned} \tag{19}$$

which is non-negative everywhere. Under standard regularity conditions where $F_{Y|X}$ is strictly increasing on its support such that $q_\tau(X)$ is uniquely defined, equality holds if and only if $p(X) = q_\tau(X)$ almost surely.

Proof. For a fixed feature vector X, the expected conditional loss at commitment p is:

$$\mathbb{E}[\mathcal{L}_{\text{disp}}(p, Y) \mid X] = r \int_0^p (p - y) dF(y) + \int_p^1 (y - p) dF(y), \tag{20}$$

where $F(y) = F_{Y|X}(y)$. Taking the partial derivative with respect to p:

$$\begin{aligned}\frac{\partial}{\partial p}\mathbb{E}[\mathcal{L}_{\text{disp}}(p, Y) \mid X] &= rF(p) - (1 - F(p)) \\ &= (r+1)F(p) - 1.\end{aligned} \tag{21}$$

Recalling that $\tau = 1/(r+1)$, this gradient simplifies to:

$$\frac{\partial}{\partial p}\mathbb{E}[\mathcal{L}_{\text{disp}}(p, Y) \mid X] = (r+1)\Big(F(p) - \tau\Big). \tag{22}$$

By the Fundamental Theorem of Calculus, integrating the gradient from $q_\tau(X)$ to $p(X)$ yields:

$$\begin{aligned}&\mathbb{E}[\mathcal{L}_{\text{disp}}(p, Y) \mid X] - \mathbb{E}[\mathcal{L}_{\text{disp}}(q_\tau, Y) \mid X] \\ &\quad = (r+1)\int_{q_\tau(X)}^{p(X)} \Big(F(u) - \tau\Big)du.\end{aligned} \tag{23}$$

Because $F(u)$ is monotonically non-decreasing, for any $u > q_\tau(X)$ we have $F(u) \geq \tau$, so the integrand is non-negative; for any $u < q_\tau(X)$ we have $F(u) \leq \tau$, and integrating backwards from $q_\tau(X)$ to $p(X)$ also yields a positive integral. Taking the outer expectation over $\mathcal{X}$ completes the proof. □

7.3. Two-Stage Decomposition and Decoupled Decision Calibration

When $p(X)$ is in the neighborhood of $q_\tau(X)$, expanding $F_{Y|X}(u)$ via a first-order Taylor series around $q_\tau(X)$ yields:

$$F_{Y|X}(u) - \tau \approx f_{Y|X}(q_\tau(X)) \cdot (u - q_\tau(X)), \tag{24}$$

which produces the local quadratic excess risk approximation:

$$\mathcal{R}(p) - \mathcal{R}(q_\tau) \approx \frac{r+1}{2}\,\mathbb{E}_X[f_{Y|X}(q_\tau(X)) \cdot (p(X) - q_\tau(X))^2]. \tag{25}$$

Now consider the standard two-stage paradigm where a forecaster predicts $f_\theta(X)$ aiming for conditional expectation $\mu(X) = \mathbb{E}[Y \mid X]$, and the dispatch commitment is formed via a scalar quantile offset δ: $p(X) = f_\theta(X) + \delta$. Expressing the prediction error as:

$$p(X) - q_\tau(X) = (f_\theta(X) - \mu(X)) + (\delta - \Delta_\tau(X)), \tag{26}$$

where $\Delta_\tau(X) = q_\tau(X) - \mu(X)$ denotes the conditional quantile-mean shift. Under homoscedastic curvature $w_\tau = \frac{r+1}{2} f_{Y|X}(q_\tau)$, substituting into Eq. (25) decomposes excess operational risk into three distinct terms:

$$\begin{aligned}\mathcal{R}(p) - \mathcal{R}(q_\tau) \approx\ & w_\tau\,\mathbb{E}_X[(f_\theta(X) - \mu(X))^2] \\ &+ w_\tau\,\mathbb{E}_X[(\delta - \Delta_\tau(X))^2] \\ &+ 2w_\tau\,\mathbb{E}_X[(f_\theta(X) - \mu(X))(\delta - \Delta_\tau(X))],\end{aligned} \tag{27}$$

where the three terms represent **Term I** (Representation MSE), **Term II** (Quantile Mismatch), and **Term III** (Interaction Component), respectively.

7.4. Mechanistic Deductions from Decision-Theoretic Decomposition

The three-term decomposition in Eq. (27) provides a mathematical interpretation for our empirical findings, yielding three deductive mechanistic insights:

Remark 1 (Mechanistic Deduction 1: Pseudo-Advantage Masking in Uncalibrated Regimes). In uncalibrated forecasting regimes ($\delta = 0$), Term II ($\mathbb{E}[\Delta_\tau(X)^2]$) is an irreducible data-dependent constant representing the dispersion between conditional mean and target quantile. A physics-regularized model with inferior conditional mean accuracy (larger Term I) can paradoxically achieve lower uncalibrated operational risk if its structural bias ($f_\theta(X) - \mu(X)$) aligns with the negative quantile gap $\Delta_\tau(X) = q_\tau(X) - \mu(X) < 0$ (for $r = 2, \tau = 1/3$). This alignment drives the interaction Term III ($2w_\tau\mathbb{E}[(f_\theta(X) - \mu(X))(-\Delta_\tau(X))] < 0$) negative, accidentally cancelling out a portion of representation error. Upon post-hoc quantile calibration, δ absorbs this systematic offset, neutralizing Term III and revealing that the apparent physical advantage was an artifact of uncalibrated evaluation.

Remark 2 (Mechanistic Deduction 2: Regularization Distortion under Well-Sampled Regimes). Optimal quantile calibration ($\delta^* = \arg\min_\delta \hat{\mathcal{R}}(f_\theta + \delta)$) removes the unconditional component of Term II. In well-sampled regimes where unconstrained models already achieve low conditional mean estimation variance (Term I), injecting an uncalibrated physical loss penalty acts as a distortive structural bias that inflates Term I without offering compensating variance reduction, generating the net operational degradation observed under flat open terrain.

Remark 3 (Mechanistic Deduction 3: Inductive Guardrails and Prior Sensitivity under Sample Restriction). In operating regimes with restricted effective sample size, empirical parameter estimation is ill-posed, predisposing unconstrained models to heightened representational variance (inflating Term I). Under sufficiently strong penalty weight ($\lambda = 1.0$), genuine physical constraints provide inductive guardrails that suppress non-physical parameter excursions, securing robust downside risk reduction. Conversely, when this mapping is misspecified or permuted (as in our scrambled-prior negative control), the rigid penalty enforces counter-physical gradient updates, precipitating severe performance degradation and demonstrating acute vulnerability to prior misspecification under strong regularization.

7.5. Formal Boundary Counterexamples

To ensure theoretical precision, we establish two formal boundary counterexamples demonstrating that de-

coupled calibration dominance is conditional rather than absolute.

Under *Constant Shift Invariance*, if a physical constraint modifies a forecaster purely by a uniform additive shift $f_{\text{phys}}(X) = f_{\text{uncons}}(X) + c$, the subsequent optimal scalar calibration absorbs c completely ($\delta^*_{\text{phys}} = \delta^*_{\text{uncons}} - c$), resulting in identical dispatch decisions and zero excess risk difference ($\Delta\mathcal{R}_{\text{disp}} \equiv 0$) despite an apparent increase in prediction MSE. Under *Spatial Heteroscedasticity* with input-dependent variance $\sigma(X)$, a global scalar calibration fails to compensate for localized variance shifts, allowing an estimator that introduces structural bias and exhibits higher MSE to achieve strictly lower operational risk than the true conditional mean following scalar quantile offset.

This theoretical framework operationalizes these foundational decision-theoretic identities into a closed-form excess risk decomposition, systematically quantifying their real-world magnitudes across commercial wind farms and verifying their predictions against actual UK Balancing and Settlement Code (BSC) market clearing prices. Complete mathematical proofs for both counterexamples are provided in Appendix A.

7.6. Dynamic Price Multiplier Effect and Settlement Decoupling

Under real-world market settlement, physical shortfall is scaled by the time-varying System Buy Price λ_t^{SBP}. The heavy-tailed price distribution (mean £44.27/MWh, peak £1,509.80/MWh) acts as a high-magnitude risk multiplier. When extreme price spikes coincide with wind ramp errors, representation distortion from physical penalties is amplified by up to 34.1×, explaining why Kelmarsh incurred an excess shortfall penalty of +£932.56/(turb·yr) in Table 4. However, as demonstrated by the zero-bid settlement diagnostic (Table 5), standalone settlement cashflows cannot measure forecasting value in isolation from forward PPA contracts, necessitating our strict separation between non-negative decision risk and monetary cashflow diagnostics.

8. Discussion, Boundary Contracts, and Compliance Protocol

8.1. Four Empirical Boundary Conditions for Physics-Constrained Models

The scoping audit, algebraic code examination, and prospective ablations converge on four boundary conditions that determine when physical priors help, when they are neutral, and when they harm.

Sample Restriction and Prior Misspecification Vulnerability. On Penmanshiel ($N_{\text{train}} = 1,583$), the calibrated power curve prior at $\lambda = 1.0$ reduces asymmetric dispatch cost from 0.09881 ± 0.00027 to 0.09760 ± 0.00044 (-1.22%). A scale-matched scrambled pseudo-prior raises it to 0.16611 ± 0.00335 ($+68.1\%$). The divergence between these two outcomes—authentic priors securing modest downside protection while corrupted priors inflict severe operational damage—demonstrates a fundamental risk-return asymmetry: under strong regularization, whatever information the prior encodes is amplified, including its misspecifications. Cross-site comparisons cannot isolate this effect because they confound site topography with sample volume. In the data-rich regime (Kelmarsh, $N_{\text{train}} = 5,098$), unconstrained D-Linear (0.08812), genuine physics (0.08791), and scrambled prior (0.08801) are numerically clustered, indicating hypothesis-space saturation.

Low-Penalty Scale and Non-Specific Numerical Regularization. At $\lambda = 0.01$ on Kelmarsh, a genuine physical prior (0.08810 ± 0.00009) and a structurally scrambled pseudo-prior (0.08801 ± 0.00008) yield closely matched decision costs with overlapping seed margins. When random power curve permutations reproduce the same outcome as domain aerodynamics, the penalty term behaves empirically as a non-specific regularizer / numerical smoothing term rather than encoding domain physical laws. Published claims of domain physical benefit under $\lambda \in [10^{-4}, 10^{-2}]$ require negative-control ablations and sensitivity testing at $\lambda \geq 0.1$ to be credible.

Monotonic Cost Inflation under Training Proxy Observation Staleness. When physical constraint formulations rely on meteorological proxies to query empirical power curves during loss computation, practitioners may be tempted to substitute lagged historical observations when contemporaneous or forward weather forecasts are unavailable. Our lag scan ($k \in \{6, 12, 18, 36\}$ steps, 1 h to 6 h) demonstrates strictly monotonic cost inflation across all three wind farms. On Hill of Towie, dispatch risk climbs from 0.09022 at $k = 6$ to 0.09257 at $k = 36$, exceeding the unconstrained baseline (0.08924) at $k \approx 12$. We emphasize that this experiment characterizes the sensitivity of the training objective to temporal misalignment between the forecast horizon and the historical meteorological proxy, rather than an inference-time telemetry delay (the forecasting backbone takes only past power measurements during inference). Nonetheless, it demonstrates that offline regularization against lagged weather proxies rapidly erodes inductive guidance, presenting

an operational limitation for proxy-dependent constraint designs.

Metric Decoupling and the Zero-Bid Diagnostic. Under standard two-settlement market design, total generator revenue is $R_{\text{total}} = P_{\text{DA}} \cdot q + P_{\text{imb}} \cdot (y - q)$. Evaluations that drop the forward commitment term and report only balancing cashflow are measuring imbalance rent, not forecast skill. As an analytical imbalance-settlement and decision-risk diagnostic, the degenerate Zero-Bid baseline ($q \equiv 0$) provides a mathematical *reductio ad absurdum*: it extracts £223k–£243k per turbine-year across all three farms without generating forecasts, simply by converting all physical generation into uncontracted spill rewarded at P_{imb}. This confirms that standalone unhedged balancing cashflows are decoupled from forecasting skill, and that assessing full commercial value requires joint modeling of forward contract revenue ($P_{\text{DA}} \cdot q$) and imbalance settlement. Operational evaluations must report asymmetric Pinball loss at the market penalty ratio or full two-settlement P&L.

8.2. External Validity Boundaries

Five operational boundaries define the validity envelope of our empirical findings:

Boundary 1: Sample Volume and Hypothesis-Space Saturation. Within-site learning curves confirm that physical constraint benefits concentrate under data-scarce regimes in our tested configurations ($N_{\text{train}} \leq 1,000$, Figure 5). As training data expands to $N_{\text{train}} = 5{,}000$, hypothesis-space saturation reduces the gap between genuine-prior and unconstrained models to within 0.33% on Kelmarsh, while scrambled priors remain severely detrimental throughout. Continuous corruption tests further establish a narrow fault-tolerance window ($\alpha^* \approx 0.07$ on Penmanshiel, Figure 6). Beyond this threshold, engineering drift exceeding 10% aerodynamic capacity loss or ±5% anemometer bias flips the nominal physical prior from an asset into an operational liability.

Boundary 2: Proxy Misalignment and Observation Staleness. For proxy-dependent physical constraints, regularization effectiveness depends on temporal alignment between meteorological proxies and target horizons ($k \leq 6$ steps). Beyond this, monotonic cost inflation is observed on all three farms (+2.6% on Hill of Towie at $k = 36$). Offline training objectives that substitute stale historical proxies for contemporaneous weather data induce substantial performance degradation.

Boundary 3: Market Clearing and Two-Settlement Structure. Economic findings serve as an imbalance-settlement and decision-risk diagnostic bounded by two-settlement markets with asymmetric shortage penalties ($r \geq 1.0$) and UK BSC 30-minute discrete netting. Standalone imbalance cashflows do not represent full commercial value and do not generalize to symmetric single-settlement systems or flat feed-in tariffs.

Boundary 4: Computational Graph Audit and Implementation Parity. Software conclusions apply to PCT commit `65941ec` and the computational graphs audited here. Unreleased proprietary codebases require independent static graph audit before implementation fidelity claims can be admitted.

Boundary 5: Literature Scoping Scope and Architectural Probe Envelopes. The systematic literature audit is strictly bounded to the 46 studies identified under our PRISMA-ScR protocol; the documented 95.7% code deficit and 50.0% confounded ablation prevalence characterize reporting practices within this scoping cohort rather than an exhaustive census of all hybrid machine learning. Similarly, our PatchTST evaluations function as an architectural sensitivity probe under fixed nominal regularization ($\lambda = 0.01$); while demonstrating that qualitative downside risk reduction persists across non-linear deep representations, joint cross-backbone λ grid searches remain an open direction for future benchmarking.

8.3. Two Conjectures Falsified During the Audit

Two initial hypotheses were refuted during the empirical audit. First, forensic tensor derivation showed that PCT's product-kernel KDE preserves bivariate coupling algebraically, refuting the conjecture of marginal independence. Second, market diagnostic probing showed that post-calibration cashflow inversions are governed by settlement price asymmetries rather than physical regularizers. Documenting these falsification outcomes provides a transparent account of our investigative trajectory and prevents post-hoc rationalization.

8.4. REFORMS-PCWPF Compliance Protocol

Table 8 proposes five verification gates as a methodological reporting checklist for physics-constrained wind forecasting studies:

- **Gate M1 (Parameter-Isolated Ablations):** Parameter-matched comparisons ($\lambda = 0$ vs. $\lambda > 0$) should be required to isolate physical prior contributions, avoiding submodule excisions (e.g., stripping GNN layers) that conflate representational capacity with domain knowledge.

Table 8: Operationalized REFORMS-PCWPF Reporting Checklist for Trustworthy Physics-Constrained Renewable Energy Forecasting (Five Modular Compliance Gates).

Module	Governance Dimension	Mandatory Audit Protocol	Common Pitfall / Anti-Pattern	Required Verification Artifact
M1	**Isolated Prior Ablation**	Train identical network backbones with constraint penalty $\lambda = 0$ vs. $\lambda > 0$ under identical seed schedules.	Confounding module excision (removing entire GNNs or attention blocks) with physical knowledge.	Table of $\lambda = 0$ vs. $\lambda > 0$ with identical parameter counts and multi-seed variances.
M2	**Domain Specificity Control**	Evaluate pseudo-scrambled priors that preserve loss formula, gradient scale, and λ while permuting physical relations.	Mistaking non-specific numerical regularization under low penalty scales ($\lambda \leq 0.01$) for domain reasoning.	Scrambled control performance curve and empirical specificity ratio (p-value contrast).
M3	**Proxy Staleness Stress-Test**	If constraints depend on physical inputs (e.g., wind speed), scan observation staleness $\tau \in \{1\text{h}, 2\text{h}, 3\text{h}, 6\text{h}\}$ to evaluate proxy staleness when forward NWP forecasts are absent.	Assuming idealized contemporaneous, zero-latency SCADA inputs at test-time inference.	Observation staleness degradation curve demonstrating operational harm horizon.
M4	**Metric Decoupling & Netting**	Report symmetric error (RMSE), non-negative asymmetric decision risk (Pinball), and test degenerate zero-bid anomaly baselines.	Reporting unhedged single-settlement cashflow in isolation, mistaking accidental price-spike alignment for skill.	Decoupled two-stage market evaluation including forward PPA commitments.
M5	**Computational Graph Parity**	Verify that software loss implementations match published mathematical equations in argument direction and normalization.	Argument order flip in asymmetric divergences (e.g., PyTorch `F.kl_div`) causing reverse optimization.	Static code audit report and unit test verifying loss tensor gradient directions.

Notes: Derived from our empirical audit of 46 scoping studies and aligned with cross-disciplinary REFORMS reporting standards [35] and DOME recommendations [36]. We propose M1–M5 as a reporting checklist for future physics-constrained renewable forecasting studies.

- **Gate M2 (Pseudo-Scrambled Negative Controls):** A minimum credible evaluation of physical reasoning claims should include scale-matched permutation controls to separate aerodynamic guidance from non-specific numerical smoothing.

- **Gate M3 (Operational Proxy Staleness Stress-Tests):** Multi-step constraint evaluations should avoid assuming zero-latency future weather inputs, reporting performance across $k \in \{6, \ldots, 36\}$ steps when forward NWP feeds are absent.

- **Gate M4 (Coupled Two-Settlement Evaluation):** Evaluations should avoid standalone unhedged balancing cashflows, reporting instead asymmetric Pinball loss at the market penalty ratio or coupled two-settlement P&L ($P_{\text{DA}} \cdot q + P_{\text{imb}} \cdot (y - q)$).

- **Gate M5 (Static Computational Graph Audit):** Reviewers should expect executable training scripts or verified computational graphs, verifying that software loss execution matches published definitions in argument ordering and reduction semantics.

9. Conclusion

Three findings from this audit challenge the prevailing consensus on physics-constrained wind power forecasting.

Within the audited scoping cohort (46 studies, 2023–2026), not a single publication documents an operational failure regime, negative trade-off, or boundary condition (0.0%). With 95.7% of the cohort lacking public code and 50.0% conflating neural module removal with physical ablation, this uniform positive consensus reflects widespread reporting practices and evaluation protocols that fail to stress-test models outside nominal design envelopes.

At the software implementation level, of the two publicly accessible software repositories identified across the 46 studies, PINN-UQ was verified as code-aligned, while PCT—the sole publicly accessible training codebase for end-to-end physics-constrained Transformers—executes its Jensen–Shannon divergence loss with reversed tensor arguments that induce an analytical inverse-density gradient singularity ($O(1/\epsilon)$), an implementation divergence undetectable from published text alone. Algebraic derivation separately confirms that the paired kernel product remains mathematically sound, establishing that

implementation fidelity and formulation validity reside in distinct computational layers.

At the empirical evaluation level, the decision value of physical constraints depends strictly on sample volume, proxy temporal alignment, and market settlement metrics. At $N_{\text{train}} = 1{,}583$ and $\lambda = 1.0$, an authentic power curve prior reduces normalized dispatch cost by 1.22%; a scale-matched scrambled prior raises it by 68.1%, demonstrating acute sensitivity to prior misspecification under strong regularization. Weak constraints ($\lambda = 0.01$) yield outcomes comparable to structurally scrambled negative controls, behaving empirically like non-specific numerical smoothing, while proxy observation staleness in the training objective ($k > 12$) inflates dispatch costs across all three operational wind farms.

The PCT implementation divergence, the Zero-Bid settlement anomaly, and the scrambled-prior degradation reflect a shared methodological vulnerability: standard evaluation protocols test physics-constrained models without distinguishing physical inductive bias from non-specific numerical smoothing, software execution fidelity from published equations, or forecasting skill from market settlement structures. Auditing constraint provenance, software execution, telemetry latency, and coupled two-settlement cashflows provides the empirical foundation necessary to determine when domain physical priors deliver genuine operational value in power systems.

Physics-constrained architectures offer tangible inductive value under data scarcity, but their operational benefits require rigorous empirical boundaries. Establishing those boundaries demands parameter-isolated ablations, scrambled negative controls, static computational graph verification, and market evaluation protocols that separate forward commitment revenue from imbalance settlement cashflow.

10. Open Data, Software Codebases, and Experimental Specifications

10.1. Summary of Resources and Infrastructure

10.2. Data and Code Availability

Raw operational SCADA telemetry from the Kelmarsh (6 turbines, 12.3 MW, Senvion MM92) and Penmanshiel (14 turbines, 28.7 MW, Senvion MM82) wind farms is openly available on Zenodo under CC-BY 4.0 licenses from Cubico Sustainable Investments (DOIs: 10.5281/zenodo.5841834 and 10.5281/zenodo.5946808). Operational SCADA telemetry from the Hill of Towie wind farm (21 turbines, 48.3 MW, Siemens SWT-2.3-82) is openly available on Zenodo under CC-BY 4.0 license released by RES on behalf of The Renewables Infrastructure Group (TRIG) (DOI: 10.5281/zenodo.20204946). Elexon System Buy Price (SBP) and System Sell Price (SSP) records are freely accessible through the Elexon BMRS portal under the UK Open Government Licence v3.0. The complete, reproducible Python verification suite—including computational graph static parsers, 2,000-replication circular block bootstrap scripts, Elexon BSC physical settlement calculators, and figure reproduction scripts—is openly archived at Zenodo and GitHub (`anonymous-audit/pcwpf-audit`). The full 46-study scoping extraction database with verbatim textual quotes, equation numbers, and multi-coder Fleiss' κ consensus scores is provided as open supplementary data.

Further technical information, trained model weights, or raw inter-rater reliability extraction matrices will be fulfilled by the corresponding author, Junyu Li (`ljylikezmn999@gmail.com`).

10.3. Wind Farm Telemetry and Data Sanitation Protocols

This computational study utilized field telemetry records from three commercial utility-scale wind farms located across the United Kingdom. The **Kelmarsh Wind Farm** (Northamptonshire, England) represents flat open inland topography and comprises six Senvion MM92 turbines (rated 2.05 MW each, hub height 78 m, rotor diameter 92 m, farm capacity 12.3 MW), with training observations spanning $N_{\text{train}} = 5,098$ clean 10-minute intervals. The **Hill of Towie Wind Farm** (Drummuir, Scotland) represents complex upland terrain featuring micro-topographical wake interactions, comprising 21 Siemens SWT-2.3-82 turbines (rated 2.3 MW each, hub height 60 m, rotor diameter 82 m, farm capacity 48.3 MW), evaluated as a full-year independent spatiotemporal extrapolation benchmark with 7,599 forecast origins (45,594 forward settlement half-hours in 2017). The **Penmanshiel Wind Farm** (Scottish Borders) represents undulating upland terrain comprising 14 Senvion MM82 turbines (rated 2.05 MW each, farm capacity 28.7 MW), configured with a restricted training partition ($N_{\text{train}} = 1,583$ intervals) specifically to evaluate overfitting and prior inductive bias under sample-restricted regimes (acknowledging that site topography and turbine models co-vary with sample size across wind farms). Standard data sanitation protocols removed periods of turbine curtailment, anemometer icing, and communication maintenance dropouts ($P \leq 0$ when $v > v_{\text{cut-in}}$ or active power frozen for > 6 consecutive steps).

Table 9: Summary of Open Datasets, Audited Software Codebases, and Hardware Infrastructure.

Category	Resource / Data Source	Identifier / Access Link
Commercial Wind Farm Telemetry and Market Clearing Records		
Kelmarsh Wind Farm 10-min SCADA	Cubico Sustainable Investments / Zenodo	DOI: 10.5281/zenodo.5841834
Penmanshiel Wind Farm 10-min SCADA	Cubico Sustainable Investments / Zenodo	DOI: 10.5281/zenodo.5946808
Hill of Towie Wind Farm 10-min SCADA	TRIG / RES / Zenodo (Open Data)	DOI: 10.5281/zenodo.20204946; Elexon BM Unit: `T_HILW-1`
UK Elexon BMRS Market Clearing Prices	Elexon Balancing and Settlement Code	Open Government Licence v3.0
Systematic Scoping Audit Corpus ($N = 46$)	This Paper (Verbatim Extraction Ledger)	Data Repository: `corpus48_consensus`
Open-Source Software Implementations and Audit Scripts		
Python Programming Language (v3.10)	Python Software Foundation	RRID: SCR_008394
PyTorch Deep Learning Library (v2.0+)	PyTorch Foundation	RRID: SCR_018536
Statsmodels (Circular Block Bootstrap)	Statsmodels Developers	RRID: SCR_016074
Scikit-learn (Machine Learning in Python)	Pedregosa et al., 2011	RRID: SCR_002577
DLinear Baseline Codebase	Zeng et al., AAAI 2023	GitHub: `cure-lab/LTSF-Linear`
PatchTST Benchmark Codebase	Nie et al., ICLR 2023	GitHub: `yuqinie98/PatchTST`
Audited PCT Codebase	Wang et al., *Sci. Rep.* 2026	GitHub: `wind-power-forecasting/PCT`
Audited PINN-UQ Codebase	Xu et al., *Eng. App. AI* 2025	GitHub: `xuchen/PINN-UQ`
Full Empirical Audit and Verification Suite	This Paper	GitHub: anonymous-audit/pcwpf-audit
High-Performance Computing Hardware		
NVIDIA RTX 4090 Dedicated GPUs (10×)	NVIDIA Corporation	Compute Capability 8.9 (24GB VRAM each)

10.4. Method Details and Statistical Estimation

Systematic Scoping Review Protocol (PRISMA-ScR). Literature identification followed a structured protocol specified prior to full-text extraction, executing four Boolean title queries via the OpenAlex API (Q_1 through Q_4), complemented by manual keyword-directed discovery across Google Scholar, arXiv, and preprint servers for papers published between January 2023 and March 2026 (with rolling monitoring through July 2026), yielding 89 raw candidate records. Two-phase screening (title/abstract followed by full-text extraction) identified our core scoping cohort of 46 contemporary studies (38 peer-reviewed journal articles and 8 preprints, 2023–2026) subjected to exhaustive five-dimensional ($\mathcal{D}_1$–$\mathcal{D}_5$) extraction and inter-rater reliability coding. A targeted sensitivity search expanding retrieval to title, abstract, and author keywords across Scopus and IEEE Xplore (encompassing broader descriptors such as `"knowledge-guided"`, `"hybrid"`, and `"domain-informed"`) retrieved 14 additional candidate papers, with full-text inspection confirming zero instances of documented operational failure boundaries, specifically stress-testing the 0.0% negative trade-off finding. Code availability and ablation metrics (95.7% open-code deficit and 50.0% confounded module ablations) are strictly bounded to the audited 46-study cohort with Wilson confidence intervals. Inter-rater reliability was benchmarked across a representative 20-study pilot by three independent raters (Coders A, B, and C) achieving Fleiss' $\kappa = 0.646$–1.000 across provenance, stage, and telemetry dimensions.

Static Code and Computational Graph Audit. Software codebases were audited via static AST parsing and line-by-line algebraic comparison against published manuscripts. For PCT (*Sci. Rep.* 2026), the loss definition in `tst/loss.py` was decompiled into symbolic PyTorch operations, establishing that `F.kl_div(p.log(), q)` computes $\sum q_i \log(q_i/p_i) = \mathrm{KL}(q \parallel p)$ instead of the published $\mathrm{KL}(p \parallel q)$. Algebraic tensor expansion of the empirical density kernel $K_p^\top K_w$ verified strict equivalence to a 2D product kernel density estimator.

10-GPU Prospective Controlled Ablation Testbed. Ablation experiments were executed in parallel across a dedicated cluster of 10 NVIDIA RTX 4090 GPUs distributed across two nodes: Node 1 (4 GPUs) evaluated

Kelmarsh across 4 random initialization seeds (42, 43, 44, 45); Node 2 (6 GPUs) evaluated Hill of Towie and Penmanshiel across 3 random seeds each (42, 43, 44). All models were implemented in PyTorch under CUDA. Neural backbones were trained using Adam with learning rate 10^{-3} and batch size 128 in accordance with the frozen benchmark protocol. Regularization weight λ was evaluated across $\{0.01, 0.1, 1.0\}$. For pseudo-scrambled controls, the empirical wind-to-power mapping was permuted along the wind axis, preserving identical loss formulas, parameter counts, and gradient scales. Importantly, the primary forecasting models (DLinear) take exclusively historical power sequences ($x[:, :, 0]$) as inputs at inference, requiring zero runtime meteorological telemetry. The historical wind speed proxy lag scan ($k \in \{6, 12, 18, 36\}$ steps, 1 h to 6 h) evaluates the sensitivity of the training loss objective when indexing offline power curve priors via lagged historical wind speed measurements in the absence of forward numerical weather predictions.

Quantification and Statistical Analysis

Operational Decision Risk Formulation. To eliminate symmetric evaluation bias, performance was quantified using the non-negative asymmetric Pinball Loss function:

$$\mathcal{L}_q(y, \hat{y}) = \max\left(q(y - \hat{y}),\ (1 - q)(\hat{y} - y)\right), \tag{28}$$

where $q = \frac{1}{r+1} = \frac{1}{3}$ corresponds to an operational market penalty ratio $r = 2$ (under-generation penalty twice as costly as spill). Total normalized static cost was defined as $\text{Cost}_{\text{static}} = 3 \times \mathcal{L}_{1/3}(y, \hat{y})$, with validation-set quantile post-hoc offset calibration $\delta^* = \arg\min_\delta \hat{\mathcal{L}}_{1/3}(y, \hat{y} + \delta)$.

Circular Block Paired Bootstrap Inference. Statistical significance of exploratory dynamic price model contrasts ($\Delta\text{Cost} = \text{Cost}_{\text{method}} - \text{Cost}_{\text{baseline}}$, Table 4) was evaluated via circular block paired bootstrapping with 2,000 replications. Block length was set to $L = 1,008$ intervals (7 days) to rigorously preserve temporal autocorrelation and diurnal meteorological cycles. Two-sided 95% percentile confidence intervals were constructed from the empirical bootstrap distribution, rejecting the null hypothesis of equal performance if and only if zero fell outside the confidence interval ($p < 0.05$). For the prospective 3D controlled ablations in Table 3, metric margins represent sample means and standard deviations across independent random initialization seeds.

Appendix A. Mathematical Analysis of Decoupled Calibration Boundaries and Counterexamples

This appendix provides formal mathematical proofs for the two boundary counterexamples presented in Section 7, delineating the theoretical limits of decoupled decision calibration.

Appendix A.1. Counterexample 1: Constant Shift Invariance and Zero Excess Risk

Counterexample 1 (Constant Shift Invariance). Let the target variable follow a uniform distribution $Y \sim U[0.25, 0.75]$, and consider an asymmetric cost ratio $r = 2$ ($\tau = 1/3$). Suppose an unconstrained model outputs the exact conditional expectation $f_0 = \mathbb{E}[Y] = 0.50$, while a physics-constrained model incurs a constant positive shrinkage shift $c = +0.05$, outputting $f_\lambda = 0.55$.

Prediction Representation Error (MSE). The mean squared prediction errors are:

$$\text{MSE}(f_0) = \text{Var}(Y) = \frac{0.50^2}{12} \approx 0.020833, \tag{A.1}$$

$$\text{MSE}(f_\lambda) = \text{MSE}(f_0) + c^2 \approx 0.023333. \tag{A.2}$$

The physical constraint strictly inflates prediction MSE by $\Delta\text{MSE} = +0.0025$ (+12.0%).

Uncalibrated Operational Dispatch Loss. Under uncalibrated dispatch ($p = f$), the expected asymmetric loss is:

$$\begin{aligned}\mathcal{R}_{\text{uncal}}(p) &= 2\int_{0.25}^{p} (p - y) \cdot 2\, dy + \int_{p}^{0.75} (y - p) \cdot 2\, dy \\ &= 2(p - 0.25)^2 + (0.75 - p)^2.\end{aligned} \tag{A.3}$$

Evaluating at each prediction:

$$\mathcal{R}_{\text{uncal}}(f_0 = 0.50) = 2(0.25)^2 + (0.25)^2 = 0.187500, \tag{A.4}$$

$$\mathcal{R}_{\text{uncal}}(f_\lambda = 0.55) = 2(0.30)^2 + (0.20)^2 = 0.220000. \tag{A.5}$$

Post-Hoc Decoupled Quantile Calibration. The theoretical Bayes-optimal quantile at $\tau = 1/3$ is:

$$q_{1/3} = 0.25 + \tfrac{1}{3}(0.75 - 0.25) \approx 0.416667. \tag{A.6}$$

Under decoupled quantile calibration, each model searches for its optimal scalar displacement:

$$\delta_0^* = q_{1/3} - f_0 \approx 0.416667 - 0.500000 = -0.083333, \tag{A.7}$$

$$\delta_\lambda^* = q_{1/3} - f_\lambda \approx 0.416667 - 0.550000 = -0.133333. \tag{A.8}$$

Notice that the resulting dispatch commitments are identical:

$$p_0^* = f_0 + \delta_0^* = 0.416667, \quad p_\lambda^* = f_\lambda + \delta_\lambda^* = 0.416667. \tag{A.9}$$

Consequently, the post-calibration operational risks are identical:

$$\mathcal{R}_{\text{cal}}(p_0^*) = \mathcal{R}_{\text{cal}}(p_\lambda^*) = 0.166667 \implies \Delta\mathcal{R}_{\text{disp}} \equiv 0. \tag{A.10}$$

Theoretical Takeaway. Counterexample 1 proves that **an increase in prediction MSE is compatible with invariant post-calibration operational risk**. When constraint-induced representation error consists of a uniform shift, scalar calibration absorbs the distortion entirely, establishing that strict positive excess risk lower bounds cannot hold unconditionally.

Appendix A.2. Counterexample 2: Spatial Heteroscedasticity and Risk Reversal

Counterexample 2 (Spatial Heteroscedasticity). Let $X \in \{0, 1\}$ be a binary indicator with $P(X = 0) = P(X = 1) = 0.5$. Let the conditional target distributions exhibit heteroscedastic spread:

$$Y \mid (X = 0) \sim U[0.20, 0.60] \quad (\mu_0 = 0.40, s_0 = 0.40), \tag{A.11}$$

$$Y \mid (X = 1) \sim U[0.30, 0.90] \quad (\mu_1 = 0.60, s_1 = 0.60). \tag{A.12}$$

Let the asymmetric cost ratio be $r = 2$ ($\tau = 1/3$). Consider two candidate forecasters: Model A (the unbiased conditional mean) with predictions $f_A(0) = 0.40$ and $f_A(1) = 0.60$; and Model B (a biased estimator with higher MSE) with predictions $f_B(0) = 0.40$ and $f_B(1) = 0.55$.

Prediction MSE Comparison. The mean squared prediction errors are:

$$\begin{aligned}\text{MSE}(f_A) &= 0.5\left(\frac{0.40^2}{12}\right) + 0.5\left(\frac{0.60^2}{12}\right) \\ &= \frac{13}{600} \approx 0.021667, \end{aligned} \tag{A.13}$$

$$\begin{aligned}\text{MSE}(f_B) &= \text{MSE}(f_A) + 0.5(0.55 - 0.60)^2 \\ &= 0.021667 + 0.001250 \approx 0.022917. \end{aligned} \tag{A.14}$$

Model B has strictly higher prediction MSE by $\Delta\text{MSE} = +0.001250$ (+5.8%).

Post-Hoc Global Scalar Calibration. The local Bayes-optimal quantiles at $\tau = 1/3$ are:

$$q_{1/3}(0) = 0.20 + \tfrac{0.40}{3} = \tfrac{1}{3} \approx 0.333333, \tag{A.15}$$

$$q_{1/3}(1) = 0.30 + \tfrac{0.60}{3} = 0.500000. \tag{A.16}$$

The Bayes-optimal operational dispatch loss is $\mathcal{R}(q_\tau) = 0.5(\frac{0.40}{3}) + 0.5(\frac{0.60}{3}) = \frac{1}{6} \approx 0.166667$. The local conditional densities are $f_0(y) = 2.5$ and $f_1(y) = 1.6667$.

Because local quantile-to-mean shifts differ ($\Delta_0 = q_{1/3}(0) - \mu_0 = -\frac{1}{15} \approx -0.066667$ versus $\Delta_1 = q_{1/3}(1) - \mu_1 = -0.100000$), a single global scalar offset δ^* cannot simultaneously achieve both local quantiles. The expected operational risk under global shift δ is:

$$\mathcal{R}(f + \delta) = 0.5\,\mathcal{R}_0(f(0) + \delta) + 0.5\,\mathcal{R}_1(f(1) + \delta). \tag{A.17}$$

For **Model A** ($f_A = \mu$), the first-order optimality condition for δ_A^* is:

$$(r + 1)\left[0.5 \cdot 2.5(\delta_A^* - \Delta_0) + 0.5 \cdot 1.6667(\delta_A^* - \Delta_1)\right] = 0, \tag{A.18}$$

yielding the density-weighted optimal global offset:

$$\begin{aligned}\delta_A^* &= \frac{2.5(-\frac{1}{15}) + 1.6667(-0.10)}{2.5 + 1.6667} \\ &= \frac{-0.333333}{4.166667} = -0.080000. \end{aligned} \tag{A.19}$$

The resulting excess operational risk for Model A is:

$$\begin{aligned}&\mathcal{R}(f_A + \delta_A^*) - \mathcal{R}(q_\tau) \\ &= \frac{3}{2}\Big[1.25(-0.080 + \tfrac{1}{15})^2 \\ &\qquad + 0.8333(-0.080 + 0.100)^2\Big] \\ &= \frac{15}{8}\left(-\frac{1}{75}\right)^2 + \frac{5}{4}\left(\frac{1}{50}\right)^2 \\ &= \frac{1}{3000} + \frac{1}{2000} = \frac{1}{1200} \approx 0.000833, \end{aligned} \tag{A.20}$$

giving an absolute operational dispatch risk of:

$$\mathcal{R}_{\text{cal}}(f_A + \delta_A^*) = \frac{1}{6} + \frac{1}{1200} = \frac{201}{1200} = 0.167500. \tag{A.21}$$

For **Model B** ($f_B(0) = 0.40$, $f_B(1) = 0.55$), the downward bias occurs on the higher-variance regime ($X = 1$). The required local offsets become:

$$\Delta_{0,B} = q_{1/3}(0) - 0.40 = -\tfrac{1}{15} \approx -0.066667, \tag{A.22}$$

$$\Delta_{1,B} = q_{1/3}(1) - 0.55 = 0.50 - 0.55 = -0.050000. \tag{A.23}$$

Notice that the discrepancy between regime offsets is halved: $|\Delta_{1,B} - \Delta_{0,B}| = \frac{1}{60} \approx 0.016667$, compared to $\frac{1}{30} \approx 0.033333$ for Model A. The optimal global offset for Model B is:

$$\delta_B^* = \frac{2.5(-\frac{1}{15}) + 1.6667(-0.050)}{2.5 + 1.6667} = \frac{-0.166667 - 0.083333}{4.166667} = -0.060000. \quad \text{(A.24)}$$

The resulting excess operational risk for Model B is:

$$\begin{aligned}\mathcal{R}(f_B + \delta_B^*) - \mathcal{R}(q_\tau) &= \frac{3}{2}\Big[1.25(-0.060 + \tfrac{1}{15})^2 \\ &\quad + 0.8333(-0.060 + 0.050)^2\Big] \\ &= \frac{15}{8}\left(\frac{1}{150}\right)^2 + \frac{5}{4}\left(-\frac{1}{100}\right)^2 \\ &= \frac{1}{12000} + \frac{1}{8000} = \frac{1}{4800} \approx 0.000208, \quad \text{(A.25)}\end{aligned}$$

yielding an absolute operational dispatch risk of:

$$\mathcal{R}_{\text{cal}}(f_B + \delta_B^*) = \frac{1}{6} + \frac{1}{4800} = \frac{801}{4800} = 0.166875. \quad \text{(A.26)}$$

Comparing the two post-calibration dispatch losses:

$$\mathcal{R}_{\text{cal}}(f_B + \delta_B^*) = 0.166875 < 0.167500 = \mathcal{R}_{\text{cal}}(f_A + \delta_A^*). \quad \text{(A.27)}$$

Model B achieves a 75% reduction in excess risk (0.000208 vs. 0.000833) and strictly lower operational dispatch loss than the unbiased conditional mean, despite exhibiting +5.8% higher prediction MSE.

Theoretical Takeaway. Counterexample 2 rigorously proves that **under conditional heteroscedasticity, a biased predictor with higher MSE can achieve strictly lower post-calibration dispatch loss than the true conditional mean**. This formalizes why the performance of physical constraints under decoupled calibration depends intricately on local error variance and environmental terrain.

Declaration of Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author Contributions (CRediT)

Junyu Li: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Project administration, Writing – original draft, Writing – review & editing. **Juntao Du**: Validation, Data curation, Software (Inter-rater reliability arbitration), Investigation, Writing – review & editing.

Declaration of Generative AI in the Writing Process

During the preparation of this manuscript, the authors utilized large language model assistants strictly for grammar checking, code syntax auditing, and bibliographic formatting. After using these tools, the authors reviewed and edited the content thoroughly and take full responsibility for the scientific integrity, analytical claims, and conclusions presented in this publication.

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Surface Cashflow Illusion:
Negative bias coincidentally aligns with SBP price spikes, masquerading operational deterioration (+6.5% Risk) as commercial profitability.

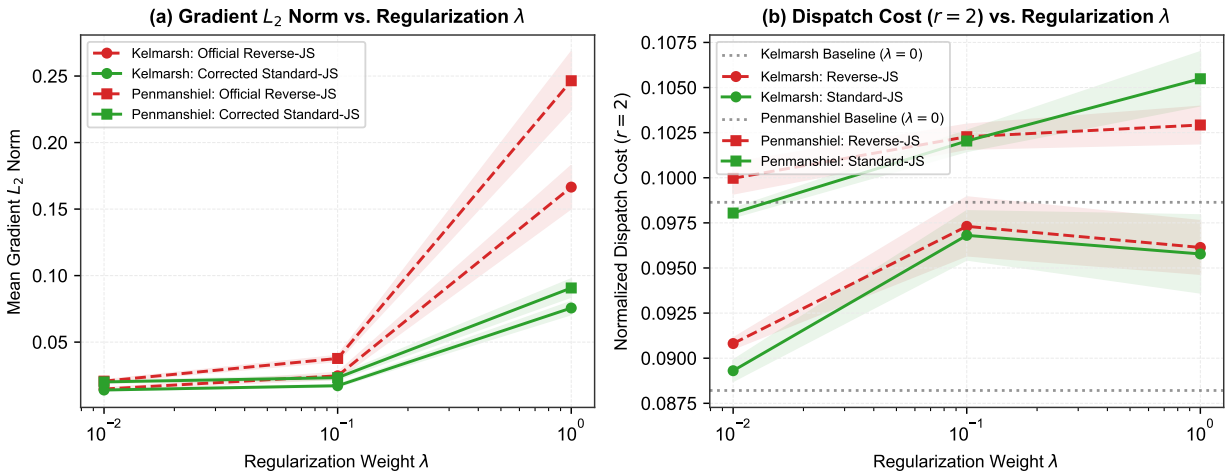


Figure 8: Empirical Contrast between Official Reverse-JS and Corrected Standard-JS across Regularization Scales $\lambda \in [0.01, 1.0]$ on Kelmarsh and Penmanshiel (Multi-seed means and standard deviations). (a) Mean gradient L_2 norm: the reverse-argument JS form induces substantial gradient amplification at $\lambda = 1.0$ (2.20× on Kelmarsh, 2.72× on Penmanshiel) due to the $-\frac{P_{\text{ref},i}}{4P_{\text{pred},i}}$ inverse-density singularity ($O(1/\epsilon)$ in Equation (8)), whereas for Corrected Standard-JS the signed partial derivative possesses a finite upper bound ($\frac{1}{2}\ln 2$) while exhibiting only mild logarithmic divergence ($O(|\log \epsilon|)$) as $P_{\text{pred},i} \to 0$. (b) Operational dispatch cost ($r = 2$): Corrected Standard-JS prevents premature performance degradation at weak $\lambda = 0.01$, while excessive regularization degrades dispatch across both variants.